

Cadi Ayyad University, Marrakech

Master's thesis — M2PA

On inhomogeneous stochastic differential equations involving local times and partial differential equations with transmission condition

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Mohamed Al Mahdi Tantaoui

Supervised by Brahim Boufoussi

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Abstract

This thesis studies one-dimensional stochastic differential equations involving the local time of the unknown process, together with the partial differential equations with transmission condition associated with them.

We first recall the construction and the main properties of the local time of a continuous semimartingale: the Tanaka and Itô–Tanaka formulas, the occupation times formula, the regularity of local times in the space variable, down-crossing approximations and symmetric local times.

We then study the time-inhomogeneous equation $X_t = x + B_t + \int_0^t \alpha(s)\, dL_s$, where L is the local time of X at 0, and prove pathwise uniqueness when $\alpha \leqslant 1/2$, following Le Gall's method; the result is extended to equations with general diffusion and drift coefficients. Next, following Le Gall, we consider equations involving the local times of the unknown process at all levels, for which we present existence and pathwise uniqueness results, limit theorems and comparison theorems.

After recalling the probabilistic representation of the solutions of the heat equation and the Feynman–Kac formula, we finally study the skew Brownian motion: its construction by random flipping of the excursions of a reflected Brownian motion, the joint law of the process and its local time, its transition density, and the parabolic partial differential equation with transmission condition that it solves.

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Introduction

The simplest stochastic differential equation involving the local time of the unknown process is

$$X_t = x + W_t + \beta \hat{L}^0_t(X),$$

where W is a Brownian motion and $\hat{L}^0(X)$ is the symmetric local time of X at 0. Its solution, the skew Brownian motion, behaves like a Brownian motion away from 0, but each excursion away from 0 is positive with probability $\frac{1+\beta}{2}$ and negative with probability $\frac{1-\beta}{2}$. Its transition semigroup solves the heat equation on each side of 0, together with a *transmission condition* at 0:

$$(1+\beta)\frac{\partial u}{\partial x}(t, 0^+) = (1-\beta)\frac{\partial u}{\partial x}(t, 0^-).$$

Equations of this kind model diffusion in media with an interface, where the coefficients of the equation are discontinuous. This thesis studies stochastic differential equations involving local times, and the parabolic partial differential equations with transmission condition associated with them.

Chapter 1 recalls the construction and the main properties of the local time of a continuous semimartingale: Tanaka's formula, the Itô–Tanaka formula, the occupation times formula, the regularity of local times in the space variable, the approximation by down-crossings, and the symmetric local time, which is used throughout the rest of the thesis.

Chapter 2 studies the time-inhomogeneous equation

$$X_t = x + B_t + \int_0^t \alpha(s)\, dL_s,$$

where L is the local time of X at 0. Following the method of Le Gall, we prove pathwise uniqueness when $\alpha \leqslant \frac{1}{2}$, and extend the result to equations of the form

$$X_t = x + \int_0^t \sigma(X_s)\, dB_s + \int_0^t b(X_s)\, ds + \int_0^t \alpha(s)\, dL_s.$$

Chapter 3 follows Le Gall [8]. It first gives pathwise uniqueness results for equations $dX_t = \sigma(t, X_t)\, dB_t + b(t, X_t)\, dt$ with non-Lipschitz coefficients. These are then used to study the singular equation

$$dX_t = \phi(X_t)\, dB_t + \int_{\mathbb{R}} \nu(da)\, d_t \hat{L}_t^a(X),$$

which involves the local times of the unknown process at all levels. The chapter ends with limit theorems and comparison theorems.

Chapter 4 recalls, following Durrett [1], how Brownian motion gives the solutions of the heat equation and of the Feynman–Kac equation.

Chapter 5 applies these ideas to the skew Brownian motion. We construct it by random flipping of the excursions of a reflected Brownian motion, compute the joint law of the process and its local time, deduce its transition density, and derive the partial differential equation with transmission condition that it solves.

About this version. This thesis was written in 2018 for the Master's degree M2PA at Cadi Ayyad University. The present version was revised in 2026: the mathematical errors of the original have been corrected, several proofs have been completed, and references have been added. The list of corrections is kept with the source files.

Chapter

1 Local Times of Semimartingales

Throughout, $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t\geqslant 0}, \mathbb{P})$ is a filtered probability space satisfying the usual conditions of right continuity and completeness. Every continuous semimartingale X is written $X = X_0 + M + A$, where M is a continuous local martingale and A a continuous adapted process of finite variation, with $M_0 = A_0 = 0$; we denote by $|A|$ the total variation process of A. The presentation of this chapter follows Durrett [1] and Revuz and Yor [13, Chapter VI].

1.1 Definition and First Properties

In this section we prove an extension of Itô's formula due to Tanaka [15] and Meyer [10]: Itô's formula remains valid, in a modified form, when f is a difference of two convex functions. Itô's formula tells us how C^2 functions act on continuous semimartingales; we now extend this to convex functions, which leads to the important notion of local time.

For a convex function $f : \mathbb{R} \to \mathbb{R}$, the left derivative

$$f'_-(x) = \lim_{h\downarrow 0} \frac{f(x) - f(x-h)}{h}$$

exists at every x, and f'_- is increasing and left continuous. Unless stated otherwise, f' denotes the left derivative f'_-.

Theorem 1.1. *Let $f : \mathbb{R} \to \mathbb{R}$ be a convex function and let X be a continuous semimartingale. Then $f(X)$ is a semimartingale and there exists a continuous adapted increasing process K^f, with $K^f_0 = 0$, such that*

$$f(X_t) - f(X_0) = \int_0^t f'_-(X_s)\, dX_s + \frac{1}{2} K^f_t.$$

Proof. Assume first that $|X_0| + |M_t| + |A|_t \leqslant N$ and $\langle M\rangle_t \leqslant N$ for all $t \geqslant 0$; in particular $|X_t| \leqslant N$.

Let g be a nonnegative C^∞ function with compact support in $(-\infty, 0]$ such that $\int_{\mathbb{R}} g(y)\,dy = 1$, and set

$$f_n(x) = \int_{\mathbb{R}} f\Big(x + \frac{y}{n}\Big) g(y)\,dy. \tag{1.1}$$

Then f_n is convex and C^∞, and Itô's formula gives

$$f_n(X_t) - f_n(X_0) = \int_0^t f_n'(X_s)\,dX_s + \frac{1}{2}K_t^n, \qquad K_t^n = \int_0^t f_n''(X_s)\,d\langle X\rangle_s. \tag{1.2}$$

Our strategy is to let $n \to \infty$ in (1.2). A convex function is Lipschitz continuous on bounded intervals, so $f_n \to f$ uniformly on compact sets and, since $|X_t| \leqslant N$,

$$\sup_t |f_n(X_t) - f(X_t)| \longrightarrow 0. \tag{1.3}$$

For the stochastic integral, differentiating (1.1) gives

$$f_n'(x) = \int_{\mathbb{R}} f_-'\Big(x + \frac{y}{n}\Big) g(y)\,dy \;\uparrow\; f_-'(x) \qquad (n \to \infty), \tag{1.4}$$

because $y \leqslant 0$ on the support of g and f_-' is increasing and left continuous. Moreover $|f_n'|$ and $|f_-'|$ are bounded on $[-N, N]$ by a constant C_N. Let

$$I_t^n = \int_0^t f_n'(X_s)\,dM_s, \qquad I_t = \int_0^t f_-'(X_s)\,dM_s.$$

Doob's L^2 maximal inequality and the isometry property of stochastic integrals give

$$\mathbb{E}\Big(\sup_t (I_t^n - I_t)^2\Big) \leqslant 4 \sup_t \mathbb{E}(I_t^n - I_t)^2 = 4\mathbb{E}\int_0^\infty (f_n'(X_s) - f_-'(X_s))^2\,d\langle M\rangle_s \longrightarrow 0$$

by the bounded convergence theorem (recall that $\langle M\rangle_\infty \leqslant N$). Passing to a subsequence (n_k),

$$\sup_t |I_t^{n_k} - I_t| \longrightarrow 0 \quad \text{almost surely.} \tag{1.5}$$

For the finite variation part, let

$$J_t^n = \int_0^t f_n'(X_s)\,dA_s, \qquad J_t = \int_0^t f_-'(X_s)\,dA_s;$$

by the bounded convergence theorem, for almost every ω,

$$\sup_t |J_t^n - J_t| \leqslant \int_0^\infty |f_n'(X_s) - f_-'(X_s)|\,d|A|_s \longrightarrow 0. \tag{1.6}$$

Finally, (1.2) can be rewritten as

$$K_t^{n_k} = 2\Big(f_{n_k}(X_t) - f_{n_k}(X_0) - \int_0^t f_{n_k}'(X_s)\,dX_s\Big). \tag{1.7}$$

By (1.3), (1.5) and (1.6), the right-hand side of (1.7) converges almost surely, uniformly in t, to $2\big(f(X_t) - f(X_0) - \int_0^t f'_-(X_s)\,dX_s\big)$. Hence so does the left-hand side; as each K^{n_k} is continuous, adapted and increasing, the limit K^f is continuous, adapted and increasing, and the formula holds.

In the general case, let $T_m = \inf\{t \geqslant 0 : |X_0| + |M_t| + |A|_t + \langle M\rangle_t \geqslant m\}$ on $\{|X_0| < m\}$. Applying the first part to the stopped process X^{T_m} on $\{|X_0| < m\}$ gives increasing processes $K^{f,m}$ such that

$$f(X_{t\wedge T_m}) = f(X_0) + \int_0^{t\wedge T_m} f'_-(X_s)\,dX_s + \frac{1}{2}K_t^{f,m}.$$

By uniqueness of the limit, $K_t^{f,m+1} = K_t^{f,m}$ for $t \leqslant T_m$, so $K_t^f = K_t^{f,m}$ for $t \leqslant T_m$ defines a continuous adapted increasing process. Since $T_m \uparrow \infty$ and $\{|X_0| < m\} \uparrow \Omega$, the formula holds for all t. □

Since $f(X_t) - f(X_0) - \int_0^t f'_-(X_s)\,dX_s$ is determined by f and X, the process K^f is unique, and $f \mapsto K^f$ is linear. If we fix X, we call K^f **the increasing process associated with** f. If f is affine, then $f(X_t) - f(X_0) = \int_0^t f'(X_s)\,dX_s$ and so $K^f = 0$.

Lemma 1.2. *For $a \in \mathbb{R}$, the increasing processes associated with $(x-a)^+$ and with $(x-a)^-$ are equal. The increasing process associated with $|x-a|$ is twice this common process.*

Proof. Let K^+ and K^- be the increasing processes associated with $(x-a)^+$ and $(x-a)^-$. Since $(x-a)^+ - (x-a)^- = x - a$ is affine, linearity gives $K^+ - K^- = K^{x-a} = 0$. Since $|x-a| = (x-a)^+ + (x-a)^-$, the process associated with $|x-a|$ is $K^+ + K^- = 2K^+$. □

Definition 1.3. The increasing process associated with $(x-a)^+$ is called the **local time of** X **at** a and is denoted by $L^a = (L_t^a)_{t\geqslant 0}$. We write $L^a(X)$ when there is a risk of ambiguity.

We define the function sgn by $\operatorname{sgn}(x) = 1$ if $x > 0$ and $\operatorname{sgn}(x) = -1$ if $x \leqslant 0$, so that the left derivative of $|x|$ is $\operatorname{sgn}(x)$.

1.1.1 Tanaka's formula

Theorem 1.4 (Tanaka's formula)**.** *Let X be a continuous semimartingale. For every $a \in \mathbb{R}$, the local time L^a is a continuous increasing process and*

$$\begin{aligned}
|X_t - a| &= |X_0 - a| + \int_0^t \operatorname{sgn}(X_s - a)\,dX_s + L_t^a,\\
(X_t - a)^+ &= (X_0 - a)^+ + \int_0^t 1_{\{X_s > a\}}\,dX_s + \frac{1}{2}L_t^a,\\
(X_t - a)^- &= (X_0 - a)^- - \int_0^t 1_{\{X_s \leqslant a\}}\,dX_s + \frac{1}{2}L_t^a.
\end{aligned}$$

In particular $|X-a|$, $(X-a)^+$ and $(X-a)^-$ are semimartingales.

Proof. The left derivatives of $(x-a)^+$, $(x-a)^-$ and $|x-a|$ are $1_{]a,\infty[}(x)$, $-1_{]-\infty,a]}(x)$ and $\operatorname{sgn}(x-a)$. The three formulas follow from Theorem 1.1 and Lemma 1.2. □

Remark 1.5. The lack of symmetry between the last two identities comes from the choice of left derivatives, which also dictates the convention $\operatorname{sgn}(0) = -1$.

Remark 1.6. The increasing process L^a defines a random Stieltjes measure dL_t^a on $\mathbb{R}_+$. It measures the "time" spent at a by the semimartingale X, as the following results show.

Theorem 1.7. *The local time L^a only increases when $X = a$: almost surely, the measure dL_t^a is carried by the set $\{t \geqslant 0 : X_t = a\}$, that is*

$$\int_0^\infty 1_{\{X_s \neq a\}}\, dL_s^a = 0.$$

Proof. The process $Y = |X-a|$ is a semimartingale with $dY_s = \operatorname{sgn}(X_s - a)\, dX_s + dL_s^a$ and $\langle Y \rangle = \langle X \rangle$. Itô's formula applied to $Y^2 = (X-a)^2$ gives

$$(X_t-a)^2 = (X_0-a)^2 + 2\int_0^t |X_s-a| \operatorname{sgn}(X_s-a)\, dX_s + 2\int_0^t |X_s-a|\, dL_s^a + \langle X \rangle_t,$$

while Itô's formula applied directly to $(X-a)^2$ gives

$$(X_t-a)^2 = (X_0-a)^2 + 2\int_0^t (X_s-a)\, dX_s + \langle X \rangle_t.$$

Since $|x-a| \operatorname{sgn}(x-a) = x-a$, comparing the two identities yields $\int_0^t |X_s-a|\, dL_s^a = 0$ for all t, almost surely, which is the claim. □

Remark 1.8. It is natural to ask whether $\{t : X_t = a\}$ is exactly the support of dL_t^a. This is not true in general, but it is true for Brownian motion [13, Chapter VI].

We are now ready for the main result of this section.

1.1.2 Itô–Tanaka formula

Theorem 1.9 (Itô–Tanaka formula)**.** *Let X be a continuous semimartingale and let f be the difference of two convex functions, with left derivative f'_- and second derivative $f'' = \mu$ in the sense of distributions (that is, μ is the signed measure such that $\mu(]x,y]) = f'_-(y+) - f'_-(x+)$). Then*

$$f(X_t) - f(X_0) = \int_0^t f'_-(X_s)\, dX_s + \frac{1}{2}\int_{\mathbb{R}} L_t^a\, \mu(da).$$

In particular, $f(X)$ is a semimartingale.

Proof. The formula is linear in f, so we may assume that f is convex and μ is a positive measure. By stopping, we may assume that $|X_t| \leqslant N$ and $\langle X \rangle_t \leqslant N$ for all t. Let

$$g(x) = \frac{1}{2} \int_{[-N,N]} |x - a| \, \mu(da).$$

Then $(f - g)'' = 0$ on $]-N, N[$ in the sense of distributions, so $f(x) - g(x) = \alpha + \beta x$ for $|x| \leqslant N$, and since the formula is trivial for affine functions, it suffices to prove it for g. The left derivative of g is

$$g'_-(x) = \frac{1}{2} \int_{[-N,N]} \operatorname{sgn}(x - a) \, \mu(da).$$

Integrating Tanaka's formula $|X_t - a| = |X_0 - a| + \int_0^t \operatorname{sgn}(X_s - a) \, dX_s + L_t^a$ with respect to $\frac{1}{2}\mu(da)$ over $[-N, N]$ gives

$$g(X_t) - g(X_0) = \frac{1}{2} \int_{[-N,N]} \Big(\int_0^t \operatorname{sgn}(X_s - a) \, dX_s \Big) \mu(da) + \frac{1}{2} \int_{[-N,N]} L_t^a \, \mu(da).$$

Lemma 1.10 and Theorem 1.11 below justify interchanging the two integrals in the first term, which gives

$$g(X_t) - g(X_0) = \int_0^t g'_-(X_s) \, dX_s + \frac{1}{2} \int_{[-N,N]} L_t^a \, \mu(da).$$

Finally, for $|a| > N$ the process X never visits a, so $L_t^a = 0$ by Theorem 1.7. This proves the result for g and completes the proof. □

For the next two results, let $(S, \mathcal{S})$ be a measurable space and μ a finite measure on $(S, \mathcal{S})$. For our purpose it would be enough to take $S = \mathbb{R}$, but it is just as easy to treat a general space. We denote by $\mathcal{P}$ the predictable σ-field on $\mathbb{R}_+ \times \Omega$, that is, the smallest σ-field containing all the sets $]s, u] \times B$ with $s < u$ and $B \in \mathcal{F}_s$, together with the sets $\{0\} \times B$ with $B \in \mathcal{F}_0$.

The first step in justifying the interchange of the order of integration is a measurability issue: given a family $H_t^a(\omega)$ of integrands indexed by $a \in S$, we want to choose the integrals $\int_0^t H_s^a \, dX_s$ so that they form a measurable function of (a, t, ω).

Lemma 1.10. *Let X be a continuous semimartingale and let $H_t^a(\omega) = H(a, t, \omega)$ be bounded and $\mathcal{S} \otimes \mathcal{P}$-measurable. Then there exists a $\mathcal{S} \otimes \mathcal{B}(\mathbb{R}_+) \otimes \mathcal{F}$-measurable function $Z(a, t, \omega)$ such that, for every $a \in S$, $(Z(a, t, \cdot))_{t \geqslant 0}$ is a continuous version of $\big(\int_0^t H_s^a \, dX_s \big)_{t \geqslant 0}$.*

Proof. By stopping, we may assume that $|A|_\infty \leqslant N$ and $\langle M \rangle_\infty \leqslant N$. Let $\mathcal{H}$ be the set of bounded $\mathcal{S} \otimes \mathcal{P}$-measurable H for which the conclusion holds. We check the assumptions of the monotone class theorem (functional version).

Clearly $\mathcal{H}$ is a vector space containing the constants. If $H(a, t, \omega) = f(a)K(t, \omega)$ where f is bounded and $\mathcal{S}$-measurable and K is bounded and $\mathcal{P}$-measurable, then $Z(a, t, \omega) = f(a) \, (K \cdot X)_t(\omega)$ is suitable, where $K \cdot X$ is a continuous version of $\int_0^{\cdot} K_s \, dX_s$. Hence $\mathcal{H}$ contains

the indicators of the sets $C \times D$ with $C \in \mathcal{S}$ and $D \in \mathcal{P}$, which form a class closed under finite intersections and generating $\mathcal{S} \otimes \mathcal{P}$.

Let now $H^n \in \mathcal{H}$ with $0 \leqslant H^n \uparrow H$, H bounded, and let Z^n be the corresponding functions. For every a, Doob's L^2 inequality, the isometry property and the bounded convergence theorem give

$$\mathbb{E}\Big(\sup_t |(H^{n,a} \cdot M)_t - (H^a \cdot M)_t|^2 \Big) \leqslant 4\,\mathbb{E} \int_0^\infty (H^{n,a}_s - H^a_s)^2 \, d\langle M \rangle_s \longrightarrow 0,$$

and, for every ω, $\sup_t |(H^{n,a} \cdot A)_t - (H^a \cdot A)_t| \leqslant \int_0^\infty |H^{n,a}_s - H^a_s| \, d|A|_s \to 0$. Thus, for every a, $Z^n(a,\cdot)$ converges to $H^a \cdot X$ uniformly in probability, and

$$\rho_n(a) = \sup_{m \geqslant n} \mathbb{E}\Big[\sup_t |Z^n(a,t) - Z^m(a,t)| \wedge 1 \Big] \longrightarrow 0 \qquad (n \to \infty).$$

Each ρ_n is $\mathcal{S}$-measurable by Fubini's theorem. Let $n_0(a) = 0$ and $n_k(a) = \inf\{n > n_{k-1}(a) : \rho_n(a) \leqslant 2^{-k}\}$; these are $\mathcal{S}$-measurable, and $\mathbb{E}\big[\sup_t |Z^{n_k(a)}(a,t) - Z^{n_{k+1}(a)}(a,t)| \wedge 1 \big] \leqslant 2^{-k}$. Hence, for every a, almost surely $\sum_k \sup_t |Z^{n_k(a)}(a,t) - Z^{n_{k+1}(a)}(a,t)| \wedge 1 < \infty$, and $Z^{n_k(a)}(a,\cdot)$ converges uniformly in t. Let $Z(a,t,\omega)$ be this limit where it exists and 0 elsewhere. Since $Z^{n_k(a)}(a,t,\omega) = \sum_n 1_{\{n_k(a)=n\}} Z^n(a,t,\omega)$ is measurable, so is Z; it is continuous in t, and for every a it is a version of $H^a \cdot X$. Thus $H \in \mathcal{H}$, and the monotone class theorem completes the proof. □

Theorem 1.11 (Fubini's theorem for stochastic integrals)**.** *Let X be a continuous semimartingale, let H^a_t be bounded and $\mathcal{S} \otimes \mathcal{P}$-measurable, and let $Z^a_t = Z(a,t,\omega)$ be any measurable function such that, for every a, Z^a is a continuous version of $\int_0^\cdot H^a_s \, dX_s$ (such a function exists by Lemma 1.10). Then $H_t = \int_S H^a_t \, \mu(da)$ is predictable and bounded, and $Y_t = \int_S Z^a_t \, \mu(da)$ is a version of $\int_0^t H_s \, dX_s$. Less formally,*

$$\int_S \Big(\int_0^t H^a_s \, dX_s \Big) \mu(da) = \int_0^t \Big(\int_S H^a_s \, \mu(da) \Big) dX_s.$$

Proof. The predictability of H follows from the classical Fubini theorem. By stopping, we may assume that $|A|_\infty \leqslant N$ and $\langle M \rangle_\infty \leqslant N$. When $X = A$ the result is the classical Fubini theorem applied for each ω, so we may assume that $X = M$. Let $\mathcal{H}$ be the set of bounded $\mathcal{S} \otimes \mathcal{P}$-measurable H for which the conclusion holds; it is a vector space containing the constants.

If $H(a,t,\omega) = f(a)K(t,\omega)$ with f bounded and $\mathcal{S}$-measurable and K bounded and $\mathcal{P}$-measurable, then $Z^a_t = f(a)(K \cdot M)_t$ and

$$\int_S Z^a_t \, \mu(da) = \Big(\int_S f(a) \, \mu(da) \Big) (K \cdot M)_t = \Big(\Big(\int_S f(a) \, \mu(da) \Big) K \Big) \cdot M_t = (H \cdot M)_t,$$

so $\mathcal{H}$ contains the indicators of the sets $C \times D$ with $C \in \mathcal{S}$ and $D \in \mathcal{P}$.

Let now $H^n \in \mathcal{H}$ with $0 \leqslant H^n \uparrow H$, H bounded. Write $Z^{n,a}$ for the versions associated with $H^{n,a}$ and $H^n_t = \int_S H^{n,a}_t \, \mu(da)$. By the Cauchy–Schwarz inequality, Doob's L^2 inequality and the

isometry property,

$$\mathbb{E}\sup_t \left| \int_S Z_t^{n,a}\,\mu(da) - \int_S Z_t^a\,\mu(da)\right| \leqslant \int_S \left(4\,\mathbb{E}\int_0^\infty (H_s^{n,a}-H_s^a)^2\,d\langle M\rangle_s\right)^{1/2}\mu(da) \longrightarrow 0$$

by the bounded convergence theorem, used twice. On the other hand $H^n \uparrow H$ boundedly, so

$$\mathbb{E}\sup_t |(H^n\cdot M)_t - (H\cdot M)_t|^2 \leqslant 4\,\mathbb{E}\int_0^\infty (H_s^n - H_s)^2\,d\langle M\rangle_s \longrightarrow 0.$$

Since $\int_S Z_t^{n,a}\,\mu(da) = (H^n\cdot M)_t$ by assumption, the two limits agree: $\int_S Z_t^a\,\mu(da) = (H\cdot M)_t$ for all t, almost surely. Hence $H\in\mathcal{H}$, and the monotone class theorem completes the proof. □

We can now complete the identification of L_t^a as the time spent at a.

1.1.3 Occupation times formula

Theorem 1.12 (Occupation times formula)**.** *Let X be a continuous semimartingale with local times (L_t^a). Almost surely, for every $t\geqslant 0$ and every bounded Borel function $g:\mathbb{R}\to\mathbb{R}$,*

$$\int_{\mathbb{R}} L_t^a\,g(a)\,da = \int_0^t g(X_s)\,d\langle X\rangle_s.$$

Proof. Let g be continuous and let $f\in C^2$ with $f''=g$. Itô's formula gives

$$f(X_t)-f(X_0) = \int_0^t f'(X_s)\,dX_s + \frac{1}{2}\int_0^t g(X_s)\,d\langle X\rangle_s,$$

while the Itô–Tanaka formula, with $\mu(da)=g(a)\,da$, gives

$$f(X_t)-f(X_0) = \int_0^t f'(X_s)\,dX_s + \frac{1}{2}\int_{\mathbb{R}} L_t^a\,g(a)\,da.$$

Comparing the two identities, $\int_{\mathbb{R}} L_t^a g(a)\,da = \int_0^t g(X_s)\,d\langle X\rangle_s$ for every continuous g, almost surely. Taking g in a countable family of continuous functions with compact support that determines measures, we obtain that, almost surely and for every t, the two measures $L_t^a\,da$ and $A\mapsto\int_0^t 1_A(X_s)\,d\langle X\rangle_s$ coincide, which is the claim. □

Remark 1.13. When X is a Brownian motion, the identity in Theorem 1.12 becomes

$$\int_{\mathbb{R}} L_t^a\,g(a)\,da = \int_0^t g(X_s)\,ds.$$

Thus L_t^a can be thought of as the time spent at a up to time t: more precisely, $a\mapsto L_t^a$ is a density of the occupation measure $\nu_t(A)=\int_0^t 1_A(X_s)\,ds$ with respect to Lebesgue measure. For a general continuous semimartingale, the occupation measure is defined with respect to $d\langle X\rangle_s$, which may be seen as the "natural" time scale of X.

There are many ways of defining the local time of Brownian motion. The one given above has the advantage of making it easy to prove the existence of a version in which L_t^a is a jointly continuous function of (a,t). This property holds for every continuous local martingale, and it is no more difficult to prove it in this generality.

Theorem 1.14. *Let X be a continuous local martingale. There is a version of the family (L_t^a) such that $(a,t) \mapsto L_t^a$ is continuous, almost surely.*

Proof. By stopping, we may assume that $|X_t| \leqslant N$ and $\langle X\rangle_t \leqslant N$ for all $t \geqslant 0$. By Tanaka's formula,

$$\frac{1}{2}L_t^a = (X_t - a)^+ - (X_0 - a)^+ - \int_0^t 1_{\{X_s > a\}}\, dX_s.$$

The map $(a,t) \mapsto (X_t - a)^+ - (X_0 - a)^+$ is continuous, so it suffices to find a jointly continuous version of

$$I_t^a = \int_0^t 1_{\{X_s > a\}}\, dX_s.$$

Fix $T < \infty$ and regard $a \mapsto I^a$ as a random variable with values in $C([0,T],\mathbb{R})$, equipped with the norm $\|f\|_\infty = \sup_{0\leqslant t\leqslant T} |f(t)|$. By Kolmogorov's continuity theorem, it suffices to show that for some constants $\alpha, \beta, C > 0$,

$$\mathbb{E}\|I^a - I^b\|_\infty^\beta \leqslant C|a-b|^{1+\alpha}.$$

Let $a < b$. Since $1_{\{X_s>a\}} - 1_{\{X_s>b\}} = 1_{\{a<X_s\leqslant b\}}$, the Burkholder–Davis–Gundy inequality, the occupation times formula and the Cauchy–Schwarz inequality give

$$\begin{aligned}\mathbb{E}\|I^a - I^b\|_\infty^4 &\leqslant C\mathbb{E}\Big(\int_0^T 1_{\{a<X_s\leqslant b\}}\, d\langle X\rangle_s\Big)^2 = C\mathbb{E}\Big(\int_a^b L_T^x\, dx\Big)^2 \\ &\leqslant C(b-a)\,\mathbb{E}\int_a^b (L_T^x)^2\, dx \leqslant C(b-a)^2 \sup_{x\in\mathbb{R}} \mathbb{E}(L_T^x)^2.\end{aligned}$$

To bound the supremum, we use Tanaka's formula

$$L_T^x = |X_T - x| - |X_0 - x| - \int_0^T \operatorname{sgn}(X_s - x)\, dX_s$$

together with $(u+v)^2 \leqslant 2u^2 + 2v^2$ and $\big||y-x| - |z-x|\big| \leqslant |y-z|$:

$$\mathbb{E}(L_T^x)^2 \leqslant 2\mathbb{E}(X_T - X_0)^2 + 2\mathbb{E}\Big(\int_0^T \operatorname{sgn}(X_s - x)\, dX_s\Big)^2 \leqslant 2(2N)^2 + 2\mathbb{E}\langle X\rangle_T \leqslant 8N^2 + 2N,$$

by the isometry property and the bounds imposed by stopping. Hence $\mathbb{E}\|I^a - I^b\|_\infty^4 \leqslant C'(b-a)^2$, which is Kolmogorov's condition with $\beta = 4$ and $\alpha = 1$. □

Example 1.15 (Reflected Brownian motion)**.** Theorem 1.14 fails for the semimartingale $X_t = |B_t|$, where B is a Brownian motion. For $a > 0$, X visits a exactly when B visits a or $-a$,

and one checks that $L_t^a(X) = L_t^a(B) + L_t^{-a}(B)$, while $L_t^a(X) = 0$ for $a < 0$. Since $L_t^0(B) > 0$ for $t > 0$, the map $a \mapsto L_t^a(X)$ is discontinuous at $a = 0$.

This example is somewhat trivial since 0 is an end point of the set of values of X. The skew Brownian motion, studied in Chapter 5, provides an example with 0 in the interior.

We now construct the regular version of local times with which we work from now on.

Theorem 1.16. *Let $X = X_0 + M + V$ be a continuous semimartingale. There exists a modification of the family $\{L_t^a; a \in \mathbb{R}, t \in \mathbb{R}_+\}$ such that, almost surely, $(a,t) \mapsto L_t^a$ is continuous in t and càdlàg in a. Moreover, for every a,*

$$L_t^a - L_t^{a-} = 2\int_0^t 1_{\{X_s=a\}}\, dV_s = 2\int_0^t 1_{\{X_s=a\}}\, dX_s.$$

In particular, if X is a local martingale, the family of local times has a bicontinuous modification.

Proof. By Tanaka's formula,

$$L_t^a = 2\Big[(X_t - a)^+ - (X_0 - a)^+ - \int_0^t 1_{\{X_s>a\}}\, dM_s - \int_0^t 1_{\{X_s>a\}}\, dV_s\Big].$$

The argument of the proof of Theorem 1.14 shows that $\widehat{M}_t^a = \int_0^t 1_{\{X_s>a\}}\, dM_s$ has a version which is jointly continuous in (a,t): after stopping so that $|X|$, $\langle M\rangle$ and $|V|$ are bounded by N, the Burkholder–Davis–Gundy inequality and the occupation times formula (recall that $\langle X\rangle = \langle M\rangle$) give $\mathbb{E}\sup_{t\leqslant T}|\widehat{M}_t^a - \widehat{M}_t^b|^4 \leqslant C(b-a)^2 \sup_x \mathbb{E}(L_T^x)^2$, and Tanaka's formula gives $\sup_x \mathbb{E}(L_T^x)^2 \leqslant 3(4N^2 + N + N^2)$. It remains to study

$$\widehat{V}_t^a = \int_0^t 1_{\{X_s>a\}}\, dV_s,$$

which is continuous in t for every a. By dominated convergence with respect to $|dV|$,

$$\lim_{b\downarrow a} \widehat{V}_t^b = \int_0^t 1_{\{X_s>a\}}\, dV_s = \widehat{V}_t^a, \qquad \lim_{b\uparrow a} \widehat{V}_t^b = \int_0^t 1_{\{X_s\geqslant a\}}\, dV_s,$$

so $a \mapsto \widehat{V}_t^a$ is càdlàg, and therefore so is $a \mapsto L_t^a$. Moreover

$$L_t^a - L_t^{a-} = 2\big(\widehat{V}_t^{a-} - \widehat{V}_t^a\big) = 2\int_0^t 1_{\{X_s=a\}}\, dV_s.$$

Finally, the occupation times formula gives

$$\int_0^t 1_{\{X_s=a\}}\, d\langle M\rangle_s = \int_0^t 1_{\{X_s=a\}}\, d\langle X\rangle_s = \int_{\mathbb{R}} 1_{\{a\}}(x) L_t^x\, dx = 0,$$

so the local martingale $\int_0^t 1_{\{X_s=a\}}\, dM_s$ has zero quadratic variation and vanishes. This gives the second equality. □

Remark 1.17. If X is a continuous process of finite variation, then $\langle X\rangle = 0$, so the occupation times formula gives $L^a(X) = 0$ for almost every a, hence for every a by right continuity.

From now on, we only consider the version of the local times given by Theorem 1.16. For this version, the following corollary gives another justification of the name "local time".

Corollary 1.18. *If X is a continuous semimartingale then, almost surely, for every a and t,*

$$L_t^a(X) = \lim_{\varepsilon\downarrow 0} \frac{1}{\varepsilon}\int_0^t 1_{[a,a+\varepsilon[}(X_s)\, d\langle X\rangle_s.$$

If M is a continuous local martingale, then almost surely, for every a and t,

$$L_t^a(M) = \lim_{\varepsilon\downarrow 0} \frac{1}{2\varepsilon}\int_0^t 1_{]a-\varepsilon,a+\varepsilon[}(M_s)\, d\langle M\rangle_s.$$

Proof. By the occupation times formula,

$$\frac{1}{\varepsilon}\int_0^t 1_{[a,a+\varepsilon[}(X_s)\, d\langle X\rangle_s = \frac{1}{\varepsilon}\int_a^{a+\varepsilon} L_t^x\, dx,$$

which converges to L_t^a as $\varepsilon \downarrow 0$ by right continuity of $x \mapsto L_t^x$. For a continuous local martingale, $x \mapsto L_t^x$ is continuous by Theorem 1.16, and $\frac{1}{2\varepsilon}\int_{a-\varepsilon}^{a+\varepsilon} L_t^x\, dx \to L_t^a$. □

Remark 1.19. For a Brownian motion B, we have in particular

$$L_t^a(B) = \lim_{\varepsilon\downarrow 0} \frac{1}{2\varepsilon}\int_0^t 1_{]a-\varepsilon,a+\varepsilon[}(B_s)\, ds,$$

which shows that $L_t^0(B)$ is measurable with respect to the completion of $\sigma(|B_s|, s \leqslant t)$.

1.1.4 Down-crossing approximations

Corollary 1.18 approximates local time by occupation times. We now give a second approximation, by the number of down-crossings. Let X be a continuous semimartingale with $X_0 = 0$. For $\varepsilon > 0$, define a double sequence of stopping times by $\sigma_0^\varepsilon = 0$, $\tau_0^\varepsilon = \inf\{t \geqslant 0 : X_t = \varepsilon\}$ and, for $n \geqslant 1$,

$$\sigma_n^\varepsilon = \inf\{t > \tau_{n-1}^\varepsilon : X_t = 0\}, \qquad \tau_n^\varepsilon = \inf\{t > \sigma_n^\varepsilon : X_t = \varepsilon\}.$$

Let $d_\varepsilon(t) = \max\{n : \sigma_n^\varepsilon < t\}$ be the number of *down-crossings* of X from level ε to level 0 before time t. In Figure 1.1, $d_\varepsilon(t) = 2$ at the right end of the picture. For simplicity, we write σ_n and τ_n for σ_n^ε and τ_n^ε, and L_t for L_t^0.

For $p \geqslant 1$, we say that $X = M + V$ belongs to $\mathcal{S}^p$ if

$$\mathbb{E}\Big[\langle M\rangle_\infty^{p/2} + \Big(\int_0^\infty |dV_s|\Big)^p\Big] < \infty.$$

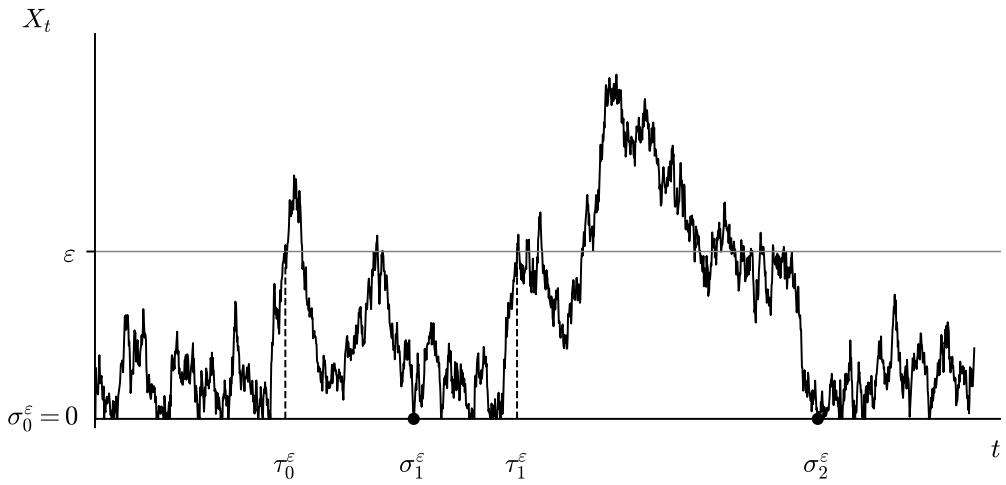


Figure 1.1: The stopping times defining the down-crossings from ε to 0, for a reflected Brownian path.

Theorem 1.20. *If $X = M + V$ belongs to $\mathcal{S}^p$ for some $p \geqslant 1$ and $X_0 = 0$, then*

$$\lim_{\varepsilon \to 0} \mathbb{E}\Big[\sup_t \Big| \varepsilon d_\varepsilon(t) - \frac{1}{2} L_t \Big|^p \Big] = 0.$$

Proof. By Tanaka's formula, for every n,

$$X^+_{\tau_n \wedge t} - X^+_{\sigma_n \wedge t} = \int_{\sigma_n \wedge t}^{\tau_n \wedge t} 1_{\{X_s > 0\}}\, dX_s + \frac{1}{2}\big(L_{\tau_n \wedge t} - L_{\sigma_n \wedge t} \big).$$

Since X does not vanish on $[\tau_n, \sigma_{n+1}[$, the local time L is constant on this interval by Theorem 1.7, and summing over n we obtain

$$\sum_{n \geqslant 0} \big(X^+_{\tau_n \wedge t} - X^+_{\sigma_n \wedge t} \big) = \int_0^t \theta^\varepsilon_s\, dX_s + \frac{1}{2} L_t, \qquad \theta^\varepsilon_s = \sum_{n \geqslant 0} 1_{]\sigma_n, \tau_n]}(s)\, 1_{]0,\varepsilon]}(X_s),$$

where θ^ε is predictable, with values in $\{0, 1\}$. The terms with $\sigma_n \geqslant t$ vanish. Each $n < d_\varepsilon(t)$ satisfies $\tau_n \leqslant t$ and contributes $X^+_{\tau_n} - X^+_{\sigma_n} = \varepsilon$, while the term $n = d_\varepsilon(t)$ contributes a quantity $u_\varepsilon(t) \in [0, \varepsilon]$. Hence the left-hand side equals $\varepsilon d_\varepsilon(t) + u_\varepsilon(t)$, and the proof will be complete once we show that

$$\lim_{\varepsilon \to 0} \mathbb{E}\Big[\sup_t \Big| \int_0^t \theta^\varepsilon_s\, dX_s \Big|^p \Big] = 0.$$

By the Burkholder–Davis–Gundy inequality, this expectation is bounded by

$$C_p \mathbb{E}\Big[\Big(\int_0^\infty \theta^\varepsilon_s\, d\langle M \rangle_s \Big)^{p/2} + \Big(\int_0^\infty \theta^\varepsilon_s\, |dV_s| \Big)^p \Big].$$

Since $\theta^\varepsilon_s \leqslant 1_{]0,\varepsilon]}(X_s) \to 0$ for every s as $\varepsilon \to 0$, the dominated convergence theorem, applied first to the inner integrals and then to the expectation (with the integrable bound provided by $X \in \mathcal{S}^p$), shows that this quantity tends to 0. □

1.2 Symmetric local times

Let X be a continuous semimartingale. Besides sgn, we use the symmetric sign function

$$\widehat{\operatorname{sgn}}(x) = 1_{\{x>0\}} - 1_{\{x<0\}}, \qquad \text{so that} \qquad \operatorname{sgn}(x) = \widehat{\operatorname{sgn}}(x) - 1_{\{x=0\}}.$$

Tanaka's formula can then be rewritten as

$$|X_t - a| = |X_0 - a| + \int_0^t \widehat{\operatorname{sgn}}(X_s - a)\, dX_s - \int_0^t 1_{\{X_s=a\}}\, dX_s + L_t^a.$$

Definition 1.21. The **symmetric local time** of X at a is

$$\hat{L}_t^a = L_t^a - \int_0^t 1_{\{X_s=a\}}\, dX_s = \frac{L_t^a + L_t^{a-}}{2},$$

where the second equality follows from Theorem 1.16.

With this definition, the symmetric Tanaka formula reads

$$|X_t - a| = |X_0 - a| + \int_0^t \widehat{\operatorname{sgn}}(X_s - a)\, dX_s + \hat{L}_t^a.$$

By Theorem 1.16, $\hat{L}^a = L^a$ when X is a local martingale, and more generally whenever $\int_0^t 1_{\{X_s=a\}}\, dV_s = 0$. The following example shows that the two local times differ in general.

Example 1.22. Let B be a Brownian motion, let $c_+, c_- > 0$ and

$$X_t = c_+ B_t^+ - c_- B_t^-.$$

By Tanaka's formula, and since $\int_0^t 1_{\{B_s=0\}}\, dB_s = 0$,

$$B_t^+ = \int_0^t 1_{\{B_s>0\}}\, dB_s + \frac{1}{2} L_t^0(B), \qquad B_t^- = -\int_0^t 1_{\{B_s<0\}}\, dB_s + \frac{1}{2} L_t^0(B),$$

hence

$$X_t = \int_0^t \left(c_+ 1_{\{B_s>0\}} + c_- 1_{\{B_s<0\}}\right) dB_s + \frac{c_+ - c_-}{2} L_t^0(B).$$

The finite variation part of X is $V = \frac{c_+ - c_-}{2} L^0(B)$. Since $\{X_s = 0\} = \{B_s = 0\}$ and $dL^0(B)$ is carried by $\{s : B_s = 0\}$ (Theorem 1.7), Theorem 1.16 gives

$$L_t^0(X) - L_t^{0-}(X) = (c_+ - c_-) \int_0^t 1_{\{B_s=0\}}\, dL_s^0(B) = (c_+ - c_-) L_t^0(B).$$

In fact, applying Tanaka's formula to $X^+ = c_+ B^+$ and $X^- = c_- B^-$ shows that $L^0(X) = c_+ L^0(B)$ and $L^{0-}(X) = c_- L^0(B)$, so that

$$\hat{L}_t^0(X) = \frac{c_+ + c_-}{2} L_t^0(B) \neq L_t^0(X) \qquad \text{when } c_+ \neq c_- \text{ and } t > 0.$$

The argument of Corollary 1.18, using that $x \mapsto L_t^x$ is càdlàg, shows that almost surely, for every a and t,

$$\hat{L}_t^a(X) = \lim_{\varepsilon \downarrow 0} \frac{1}{2\varepsilon} \int_0^t 1_{]a-\varepsilon, a+\varepsilon[}(X_s)\, d\langle X \rangle_s.$$

We now state the Itô–Tanaka formula in terms of symmetric local times.

Theorem 1.23 (Symmetric Itô–Tanaka formula)**.** *Let X be a continuous semimartingale and let f be the difference of two convex functions, with symmetric derivative $f'_{\mathrm{s}} = \frac{1}{2}(f'_- + f'_+)$ and with $f'' = \mu$ in the sense of distributions. Then*

$$f(X_t) - f(X_0) = \int_0^t f'_{\mathrm{s}}(X_s)\, dX_s + \frac{1}{2} \int_{\mathbb{R}} \hat{L}_t^a\, \mu(da).$$

Proof. The proof is the same as that of Theorem 1.9, with sgn replaced by $\widehat{\mathrm{sgn}}$ and Tanaka's formula replaced by its symmetric version. With $g(x) = \frac{1}{2} \int_{[-N,N]} |x-a|\, \mu(da)$, the symmetric derivative of g is $g'_{\mathrm{s}}(x) = \frac{1}{2} \int_{[-N,N]} \widehat{\mathrm{sgn}}(x-a)\, \mu(da)$, since the symmetric derivative of $|x-a|$ is $\widehat{\mathrm{sgn}}(x-a)$. Integrating the symmetric Tanaka formula with respect to $\frac{1}{2}\mu(da)$ and applying Theorem 1.11 gives

$$g(X_t) - g(X_0) = \int_0^t g'_{\mathrm{s}}(X_s)\, dX_s + \frac{1}{2} \int_{[-N,N]} \hat{L}_t^a\, \mu(da),$$

and $\hat{L}_t^a = 0$ for $|a| > N$, which completes the proof. □

Chapter

2 SDEs Involving the Local Time at Zero

In this chapter we study the one-dimensional stochastic differential equation

$$X_t = x + B_t + \int_0^t \alpha(s)\, dL_s, \tag{2.1}$$

where $x \in \mathbb{R}$, B is a Brownian motion started at 0, $L = L^0(X)$ is the (right-continuous) local time at 0 of the unknown semimartingale X, as defined in Chapter 1, and $\alpha : \mathbb{R}_+ \to \mathbb{R}$ is a deterministic, locally bounded Borel function. The coefficient of the local time depends on time, which is why the equation is called *inhomogeneous*.

A solution of (2.1) is a pair (X, B) of continuous adapted processes on a filtered probability space satisfying the usual conditions, such that B is an $(\mathcal{F}_t)$-Brownian motion and (2.1) holds almost surely for all t. We say that *uniqueness in law* holds if any two solutions, possibly defined on different spaces, have the same law, and that *pathwise uniqueness* holds if any two solutions defined on the same space with the same Brownian motion are indistinguishable.

Theorem 2.1. *If $\alpha(s) \leqslant \frac{1}{2}$ for every $s \geqslant 0$, pathwise uniqueness holds for (2.1).*

Remark 2.2. Let X be a solution of (2.1) and let $\hat{L} = \hat{L}^0(X)$ be its symmetric local time at 0. Since $X = x + B + V$ with $V = \int_0^{\cdot} \alpha(s)\, dL_s$, and dL is carried by $\{s : X_s = 0\}$, Theorem 1.16 gives

$$L_t - L_t^{0-} = 2\int_0^t 1_{\{X_s=0\}}\, \alpha(s)\, dL_s = 2\int_0^t \alpha(s)\, dL_s, \qquad \hat{L}_t = \int_0^t (1-\alpha(s))\, dL_s. \tag{2.2}$$

When $\alpha(s) = \alpha < 1$ is constant, (2.1) can therefore be written $X_t = x + B_t + \frac{\alpha}{1-\alpha}\hat{L}_t$, and X is the skew Brownian motion of parameter $\beta = \frac{\alpha}{1-\alpha}$ studied in Chapter 5. The condition $\alpha \leqslant \frac{1}{2}$ corresponds exactly to $\beta \in]-1, 1]$, the value $\beta = 1$ (reflected Brownian motion) being obtained for $\alpha = \frac{1}{2}$.

To prove Theorem 2.1 we follow the method of Le Gall [7], itself inspired by Perkins [11]: we first prove uniqueness in law for (2.1), and then show that if X^1 and X^2 are two solutions with the same Brownian motion, then $X^1 \vee X^2$ is also a solution. These two facts imply pathwise uniqueness. We start with a deterministic lemma due to Skorokhod [13, Lemma VI.2.1].

Lemma 2.3 (Skorokhod's lemma)**.** *Let y be a continuous real function on $[0,\infty[$ with $y(0) \geqslant 0$. There exists a unique pair (z,l) of continuous functions on $[0,\infty[$ such that*

(i) $z = y + l$;

(ii) $z \geqslant 0$;

(iii) l *is increasing,* $l(0) = 0$*, and the measure* dl *is carried by* $\{u : z(u) = 0\}$*, that is,* $\int_0^\infty 1_{\{z(u)>0\}}\, dl(u) = 0$.

This pair is given by

$$l(u) = \sup_{0\leqslant s\leqslant u} \big(-y(s)\big) \vee 0, \qquad z(u) = y(u) + l(u).$$

Proof. Existence. The function l defined above is continuous, increasing, $l(0) = 0$, and $z = y+l \geqslant 0$ since $l \geqslant -y$. Let t be such that $z(t) > 0$, that is $-y(t) < l(t)$. By continuity there is $\delta > 0$ such that $-y(s) < l(t)$ for $|s-t| \leqslant \delta$. For $u \in [t, t+\delta]$, we get $l(u) = l(t) \vee \sup_{t\leqslant s\leqslant u}(-y(s) \vee 0) = l(t)$. If $l(t) > 0$, the same argument on $[t-\delta, t]$ gives $\sup_{t-\delta\leqslant s\leqslant t}(-y(s)\vee 0) < l(t)$, hence $l(t-\delta) = l(t)$; if $l(t) = 0$, then $l = 0$ on $[0,t]$. In both cases l is constant on a neighbourhood of t. The open set $\{z > 0\}$ is a countable union of such neighbourhoods, so $dl(\{z > 0\}) = 0$.

Uniqueness. Let (z_1, l_1) and (z_2, l_2) be two solutions. Then $z_1 - z_2 = l_1 - l_2$ is continuous and of finite variation, so by the integration by parts formula for such functions,

$$\begin{aligned} 0 \leqslant (z_1(u) - z_2(u))^2 &= 2\int_0^u (z_1(v) - z_2(v))\, d(l_1 - l_2)(v) \\ &= -2\int_0^u z_1(v)\, dl_2(v) - 2\int_0^u z_2(v)\, dl_1(v) \leqslant 0, \end{aligned}$$

where we used $\int_0^u z_i\, dl_i = 0$ by (iii), and $z_j \geqslant 0$, $dl_i \geqslant 0$. Hence $z_1 = z_2$ and $l_1 = l_2$. □

An important application of Skorokhod's lemma is the following theorem of Lévy. Let $S_t = \sup_{s\leqslant t} B_s$.

Theorem 2.4 (Lévy's theorem)**.** *The following identity in law holds:*

$$\big(|B_t|, L_t^0(B); t \geqslant 0\big) \overset{(law)}{=} \big(S_t - B_t, S_t; t \geqslant 0\big).$$

Proof. By Tanaka's formula, $|B_t| = \beta_t + L_t^0(B)$, where $\beta_t = \int_0^t \operatorname{sgn}(B_s)\, dB_s$ is a Brownian motion by Lévy's characterisation. By Lemma 2.3 applied to $y = \beta$, $L_t^0(B) = \sup_{s\leqslant t}(-\beta_s)$. Hence, with $W = -\beta$,

$$\Big(|B_t|, L_t^0(B)\Big) = \Big(\sup_{s\leqslant t} W_s - W_t, \sup_{s\leqslant t} W_s\Big),$$

and the result follows since W and B have the same law. □

The same argument gives the following characterisation of reflected Brownian motion.

Proposition 2.5. *Let $z \geqslant 0$ and let B be a Brownian motion started at 0. Let X and K be continuous processes such that*

(i) $X_t = z - B_t + K_t \geqslant 0$ *for all t;*

(ii) $K_0 = 0$ *and $t \mapsto K_t$ is increasing;*

(iii) $\int_0^\infty 1_{]0,+\infty[}(X_s)\, dK_s = 0$.

Then X has the same law as $|W|$, where W is a Brownian motion started at z.

Proof. By Lemma 2.3, $X = \Phi(z - B)$, where Φ maps y to the function z of the lemma. By Tanaka's formula, $|W_t| = z + \gamma_t + L_t^0(W)$ with $\gamma_t = \int_0^t \operatorname{sgn}(W_s)\, dW_s$ a Brownian motion, so $|W| = \Phi(z + \gamma)$ by the same lemma. Since $-B$ and γ are Brownian motions started at 0, X and $|W|$ have the same law. □

We now state the two general facts on which the method relies. The first one requires the following extension of the occupation times formula.

Lemma 2.6. *Let X be a continuous semimartingale. Almost surely, for every bounded measurable process H (measurable in (s,ω)) and every bounded Borel function g,*

$$\int_0^t H_s\, g(X_s)\, d\langle X\rangle_s = \int_{\mathbb{R}} g(a)\Big(\int_0^t H_s\, dL_s^a\Big) da.$$

Proof. For $H = 1_{]u,v]}$ this is the difference of the occupation times formula (Theorem 1.12) at times $t \wedge v$ and $t \wedge u$. The general case follows by the monotone class theorem. □

Lemma 2.7. *Let f be a continuous function of finite variation on $\mathbb{R}_+$. Then the measure $1_{\{f(s)=0\}}\, df(s)$ is zero.*

Proof. Let g_n be continuous functions with $0 \leqslant g_n \leqslant 1$ and $g_n \downarrow 1_{\{0\}}$, and let $G_n(y) = \int_0^y g_n(z)\, dz$. By the chain rule for continuous functions of finite variation, $\int_u^t g_n(f(s))\, df(s) = G_n(f(t)) - G_n(f(u))$, which tends to 0 since $|G_n(y)| \leqslant \int_{-|y|}^{|y|} g_n \to 0$. By dominated convergence with respect to $|df|$, $\int_u^t 1_{\{f(s)=0\}}\, df(s) = 0$ for all $u < t$. □

In the next lemma, $L^{0-}(Z) = \lim_{a\uparrow 0} L^a(Z)$ denotes the left local time at 0 of a continuous semimartingale Z.

Lemma 2.8. *Let X^1 and X^2 be continuous semimartingales such that $\Delta = X^1 - X^2$ has finite variation. Let $L^i = L^0(X^i)$, $L^{i,-} = L^{0-}(X^i)$, let V^1 be the finite variation part of X^1, and let*

$$Y = X^1 \vee X^2, \qquad U = X^1 - (X^2)^+ = X^1 \wedge \Delta.$$

Then, as measures on $\mathbb{R}_+$,

(a) $dL^0_s(Y) = dL^0_s(U) + 1_{\{X^1_s \leqslant 0\}}\, dL^2_s$;

(b) $1_{\{\Delta_s>0\}}\, dL^1_s \leqslant dL^0_s(U) \leqslant 1_{\{\Delta_s\geqslant 0\}}\, dL^1_s$ *and* $1_{\{\Delta_s>0\}}\, dL^{1,-}_s \leqslant dL^{0-}_s(U) \leqslant 1_{\{\Delta_s\geqslant 0\}}\, dL^{1,-}_s$;

(c) $dL^0_s(U) - dL^{0-}_s(U) = 2\cdot 1_{\{X^1_s=0,\, X^2_s\leqslant 0\}}\, dV^1_s - 1_{\{X^1_s = X^2_s = 0\}}\, dL^2_s$.

Proof. (a) Since Δ has finite variation, its local time vanishes, and Tanaka's formula gives $Y = X^2 + \Delta^+ = X^2 + \int_0^{\cdot} 1_{\{\Delta_s>0\}}\, d\Delta_s$, that is $dY_s = 1_{\{\Delta_s>0\}}\, dX^1_s + 1_{\{\Delta_s\leqslant 0\}}\, dX^2_s$. By Tanaka's formula,

$$dY^+_s = 1_{\{X^1_s>0,\,\Delta_s>0\}}\, dX^1_s + 1_{\{X^2_s>0,\,\Delta_s\leqslant 0\}}\, dX^2_s + \frac{1}{2}\, dL^0_s(Y). \tag{2.3}$$

On the other hand, $Y^+ = (X^1)^+ \vee (X^2)^+ = (X^2)^+ + U^+$; indeed, if $X^1 > 0$ then $\big((X^1)^+ - (X^2)^+\big)^+ = U^+$, and if $X^1 \leqslant 0$ both sides vanish. Now $dU_s = 1_{\{X^2_s\leqslant 0\}}\, dX^1_s + 1_{\{X^2_s>0\}}\, d\Delta_s - \frac{1}{2}\, dL^2_s$, and $\{U>0\} = \{X^1>0,\, \Delta>0\}$. Since dL^2 is carried by $\{X^2=0\}$, on which $U = X^1$, Tanaka's formula for $(X^2)^+$ and U^+ gives

$$dY^+_s = 1_{\{X^1_s>0,\,\Delta_s>0\}}\, dX^1_s + 1_{\{X^2_s>0,\,\Delta_s\leqslant 0\}}\, dX^2_s + \frac{1}{2}\, 1_{\{X^1_s\leqslant 0\}}\, dL^2_s + \frac{1}{2}\, dL^0_s(U). \tag{2.4}$$

Comparing (2.3) and (2.4) gives (a).

(b) The martingale part of U is $\int 1_{\{X^2\leqslant 0\}}\, dM$, where M is the common martingale part of X^1 and X^2, and $U = X^1$ on $\{X^2\leqslant 0\}$. By Lemma 2.6, for every bounded Borel g,

$$\int_0^t g(U_s)\, d\langle U\rangle_s = \int_0^t g(X^1_s) 1_{\{X^2_s\leqslant 0\}}\, d\langle X^1\rangle_s = \int_{\mathbb{R}} g(b)\Big(\int_0^t 1_{\{X^2_s\leqslant 0\}}\, dL^b_s(X^1)\Big) db.$$

Since $dL^b(X^1)$ is carried by $\{X^1 = b\}$, on which $X^2\leqslant 0$ if and only if $\Delta\geqslant b$, we get, for almost every b and as measures, $dL^b_s(U) = 1_{\{\Delta_s\geqslant b\}}\, dL^b_s(X^1)$. As $b\downarrow 0$ (resp. $b\uparrow 0$) outside a null set, $L^b_s(U)\to L^0_s(U)$ (resp. $L^{0-}_s(U)$), and $L^b_s(X^1)\to L^1_s$ (resp. $L^{1,-}_s$) for every s, with continuous limits; hence the corresponding measures on any interval $[u,t]$ converge weakly. For $0<b<\delta$, the set $\{\Delta\geqslant b\}$ contains the open set $\{\Delta>\delta\}$ and is contained in the closed set $\{\Delta\geqslant 0\}$, so by the portmanteau theorem

$$1_{\{\Delta>\delta\}}\, dL^1 \leqslant dL^0(U) \leqslant 1_{\{\Delta\geqslant 0\}}\, dL^1,$$

and letting $\delta \downarrow 0$ gives the first pair of inequalities. For $b < 0$, the set $\{\Delta \geqslant b\}$ contains the open set $\{\Delta > 0\}$ and, for $b'' \leqslant b$, is contained in the closed set $\{\Delta \geqslant b''\}$; letting $b \uparrow 0$ and then $b'' \uparrow 0$ gives the second pair.

(c) By Theorem 1.16, $dL^0(U) - dL^{0-}(U) = 2 \cdot 1_{\{U=0\}}\, dV^U$, where $dV^U = 1_{\{X^2 \leqslant 0\}}\, dV^1 + 1_{\{X^2>0\}}\, d\Delta - \frac{1}{2}\, dL^2$ is the finite variation part of U. On $\{X^2 \leqslant 0\}$ we have $U = X^1$; on $\{X^2 > 0\}$ we have $U = \Delta$ and $1_{\{\Delta=0\}}\, d\Delta = 0$ by Lemma 2.7; and on the support of dL^2 we have $U = X^1$. This gives (c). □

Proposition 2.9. *Let X^1 and X^2 be two solutions of (2.1), defined on the same space with the same Brownian motion B and the same initial value x. Then $X^1 - X^2$ has finite variation, and $X^1 \vee X^2$ is also a solution of (2.1).*

Proof. The difference $\Delta = X^1 - X^2 = \int_0^{\cdot} \alpha(s)\, d(L^1_s - L^2_s)$ has finite variation, so its local time vanishes, and as in the proof of Lemma 2.8(a),

$$X^1 \vee X^2 = X^2 + \int_0^{\cdot} 1_{\{\Delta_s>0\}}\, d\Delta_s.$$

Since dL^1 is carried by $\{X^1 = 0\}$, on which $\Delta > 0$ if and only if $X^2 < 0$, and dL^2 is carried by $\{X^2 = 0\}$, on which $\Delta \leqslant 0$ if and only if $X^1 \leqslant 0$, we get

$$X^1_t \vee X^2_t = x + B_t + \int_0^t \alpha(s)\, d\Lambda_s, \qquad \Lambda_t = \int_0^t 1_{\{X^2_s<0\}}\, dL^1_s + \int_0^t 1_{\{X^1_s\leqslant 0\}}\, dL^2_s.$$

It remains to show that $\int_0^t \alpha(s)\, dL^0_s(X^1 \vee X^2) = \int_0^t \alpha(s)\, d\Lambda_s$. Let $T_0 = \{s : X^1_s = X^2_s = 0\}$. By Lemma 2.8(a) and (b), $dL^0(X^1 \vee X^2) = d\Lambda + d\theta$, where θ is increasing and $0 \leqslant d\theta \leqslant 1_{T_0}\, dL^1$; similarly $dL^{0-}(U) = 1_{\{\Delta>0\}}\, dL^{1,-} + d\theta^-$ with $0 \leqslant d\theta^- \leqslant 1_{T_0}\, dL^{1,-}$. By (2.2), $dL^{1,-} = (1-2\alpha)\, dL^1$, and $dV^1 = \alpha\, dL^1$. Inserting this in Lemma 2.8(c) gives

$$d\theta - d\theta^- = 2\alpha\, 1_{T_0}\, dL^1 - 1_{T_0}\, dL^2.$$

By Lemma 2.7 applied to Δ, $1_{\{\Delta=0\}}\, \alpha\, (dL^1 - dL^2) = 0$, so $dL^1 = dL^2$ on $T_0 \cap \{\alpha \neq 0\}$. There,

$$d\theta = d\theta^- - (1-2\alpha)\, dL^1 \leqslant (1-2\alpha)\, dL^1 - (1-2\alpha)\, dL^1 = 0,$$

hence $1_{\{\alpha\neq 0\}}\, d\theta = 0$ and $\int_0^t \alpha(s)\, d\theta_s = 0$. Therefore $X^1 \vee X^2 = x + B + \int_0^{\cdot} \alpha(s)\, dL^0_s(X^1 \vee X^2)$. □

Proposition 2.10. *If uniqueness in law holds for (2.1), then pathwise uniqueness holds for (2.1).*

Proof. Let X^1 and X^2 be two solutions with the same Brownian motion and initial value. By Proposition 2.9, $Y = X^1 \vee X^2$ is a solution, so Y and X^1 have the same law. Since $Y \geqslant X^1$, for every t, $\mathbb{E}[\arctan Y_t - \arctan X^1_t] = 0$ with a nonnegative integrand, hence $Y_t = X^1_t$ almost surely. By continuity, Y and X^1 are indistinguishable, and similarly for X^2. Hence $X^1 = X^2$. □

Proof of Theorem 2.1. By Proposition 2.10, it suffices to prove uniqueness in law. Let X be a solution, $L = L^0(X)$ and $\hat{L} = \hat{L}^0(X)$.

Step 1: the law of $(|X|, L)$ is determined. By the occupation times formula, $\int_0^t 1_{\{X_s=0\}}\, ds = 0$. The process $\beta_t = \int_0^t \widehat{\operatorname{sgn}}(X_s)\, dB_s$ is therefore a continuous local martingale with $\langle\beta\rangle_t = t$, hence a Brownian motion. By the symmetric Tanaka formula, and since $\widehat{\operatorname{sgn}}(0) = 0$ and dL is carried by $\{X = 0\}$,

$$|X_t| = |x| + \int_0^t \widehat{\operatorname{sgn}}(X_s)\, dX_s + \hat{L}_t = |x| + \beta_t + \hat{L}_t.$$

The process $\hat{L}$ is continuous, increasing, and $d\hat{L}$ is carried by $\{|X| = 0\}$, so by Lemma 2.3, $\hat{L}_t = \sup_{s\leqslant t}(-|x| - \beta_s) \vee 0$, and $(|X|, \hat{L})$ is a deterministic functional of the Brownian motion β. Finally, by (2.2) and since $1 - \alpha(s) \geqslant \frac{1}{2}$,

$$L_t = \int_0^t \frac{d\hat{L}_s}{1 - \alpha(s)}.$$

Hence the law of $(|X|, L)$ is uniquely determined. In particular $\mathbb{E}L_t \leqslant 2\mathbb{E}\hat{L}_t \leqslant 2\mathbb{E}\sup_{s\leqslant t}|\beta_s| < \infty$.

Step 2: the law of X_t is determined for every t. For $\lambda \in \mathbb{R}$, let $g_t(\lambda) = \mathbb{E}\exp(i\lambda X_t)$. By Itô's formula, and since $e^{i\lambda X_s} = 1$ on the support of dL,

$$e^{i\lambda X_t} = e^{i\lambda x} + i\lambda \int_0^t e^{i\lambda X_s}\, dB_s + i\lambda \int_0^t \alpha(s)\, dL_s - \frac{\lambda^2}{2}\int_0^t e^{i\lambda X_s}\, ds.$$

The stochastic integral is a martingale, so taking expectations,

$$g_t(\lambda) = e^{i\lambda x} - \frac{\lambda^2}{2}\int_0^t g_s(\lambda)\, ds + i\lambda h(t), \qquad h(t) = \mathbb{E}\Big[\int_0^t \alpha(s)\, dL_s\Big].$$

By Step 1, h is uniquely determined, and this linear integral equation has a unique solution by Gronwall's lemma, namely $g_t(\lambda) = e^{-\lambda^2 t/2}e^{i\lambda x} + i\lambda\int_0^t e^{-\lambda^2(t-s)/2}\, dh(s)$. Hence the law of X_t is uniquely determined, for every initial value x and every function α satisfying the assumptions.

Step 3: the finite-dimensional distributions are determined. Fix $s \geqslant 0$. Under a regular conditional distribution of the solution given $\mathcal{F}_s$, the process $(X_{s+u})_{u\geqslant 0}$ is, for almost every ω, a solution of (2.1) started at $X_s(\omega)$, with α replaced by $\alpha(s + \cdot)$ and driven by the Brownian motion $(B_{s+u} - B_s)_{u\geqslant 0}$; this uses that $L_{s+u} - L_s = L^0_u\big(X_{s+\cdot}\big)$. This is the classical argument of Stroock and Varadhan, see Karatzas and Shreve [6, Section 5.4]. By Step 2, the conditional law of X_{s+u} given $\mathcal{F}_s$ is therefore a kernel $Q_{s,s+u}(X_s, \cdot)$ which does not depend on the solution. By induction on the number of times, the finite-dimensional distributions of X are uniquely determined, and so is its law. □

Remark 2.11. The condition $\alpha \leqslant \frac{1}{2}$ is essential. By (2.2), the left local time of a solution is $L_t^{0-} = \int_0^t (1 - 2\alpha(s))\, dL_s$, which must be increasing. Suppose for instance that $\alpha(s) > \frac{1}{2}$ for all

s. Then L^{0-} is both nonnegative and decreasing with $L^{0-}_0 = 0$, so $L^{0-} = 0$, and therefore $L = 0$ since $1 - 2\alpha < 0$. Then $X = x + B$, whose local time at 0 is not identically zero, a contradiction: (2.1) has no solution.

We now extend Theorem 2.1 to equations with general diffusion and drift coefficients.

Theorem 2.12. *Let α be as in Theorem 2.1, and let σ and b be bounded Borel functions on $\mathbb{R}$ such that σ is Lipschitz continuous and $\sigma \geqslant \varepsilon$ for some constant $\varepsilon > 0$. Then pathwise uniqueness holds for the equation*

$$X_t = x + \int_0^t \sigma(X_s)\, dB_s + \int_0^t b(X_s)\, ds + \int_0^t \alpha(s)\, dL^0_s(X). \tag{2.5}$$

Proof. We reduce to the case $\sigma = 1$. Let $h(x) = \int_0^x \frac{dy}{\sigma(y)}$. Since $1/\sigma$ is Lipschitz and takes values in $[1/\|\sigma\|_\infty, 1/\varepsilon]$, h is a strictly increasing C^1 bijection of $\mathbb{R}$ with $h(0) = 0$, and h' is Lipschitz, so that h'' exists in the sense of distributions and is a bounded function $\phi = -\sigma'/\sigma^2$.

Let X be a solution of (2.5), $L = L^0(X)$ and $Y = h(X)$. By the Itô–Tanaka formula and the occupation times formula, and since $X = 0$ on the support of dL,

$$Y_t = h(x) + B_t + \int_0^t \left(\frac{b}{\sigma} + \frac{1}{2}\sigma^2\phi\right)(X_s)\, ds + \int_0^t \frac{\alpha(s)}{\sigma(0)}\, dL_s.$$

We now compute the local time of Y. Since $d\langle Y\rangle_s = ds$ and $d\langle X\rangle_s = \sigma^2(X_s)\, ds$, Corollary 1.18 and the occupation times formula give

$$L^0_t(Y) = \lim_{\varepsilon\downarrow 0} \frac{1}{\varepsilon}\int_0^t 1_{\{0\leqslant X_s < h^{-1}(\varepsilon)\}} \frac{d\langle X\rangle_s}{\sigma^2(X_s)} = \lim_{\varepsilon\downarrow 0} \frac{1}{\varepsilon}\int_0^{h^{-1}(\varepsilon)} \frac{L^a_t}{\sigma^2(a)}\, da = \frac{L_t}{\sigma(0)},$$

by right continuity of $a \mapsto L^a_t$ and since $(h^{-1})'(0) = \sigma(0)$. Hence Y solves

$$Y_t = h(x) + B_t + \int_0^t \hat{b}(Y_s)\, ds + \int_0^t \alpha(s)\, dL^0_s(Y), \qquad \hat{b} = \left(\frac{b}{\sigma} + \frac{1}{2}\sigma^2\phi\right) \circ h^{-1}, \tag{2.6}$$

and conversely $X = h^{-1}(Y)$ solves (2.5) if Y solves (2.6). It therefore suffices to prove pathwise uniqueness for (2.6), whose drift $\hat{b}$ is bounded and Borel.

Uniqueness in law. Let Y be a solution of (2.6) and $T > 0$. By Girsanov's theorem, under the probability $\mathbb{Q}$ with density $\exp\left(-\int_0^T \hat{b}(Y_s)\, dB_s - \frac{1}{2}\int_0^T \hat{b}^2(Y_s)\, ds\right)$, the process $W = B + \int_0^\cdot \hat{b}(Y_s)\, ds$ is a Brownian motion on $[0, T]$, and $Y = h(x) + W + \int_0^\cdot \alpha(s)\, dL^0_s(Y)$. By the proof of Theorem 2.1, the law of Y under $\mathbb{Q}$ is uniquely determined. Since W is a functional of Y, the density $d\mathbb{P}/d\mathbb{Q} = \exp\left(\int_0^T \hat{b}(Y_s)\, dW_s - \frac{1}{2}\int_0^T \hat{b}^2(Y_s)\, ds\right)$ is a functional of Y, so the law of Y under $\mathbb{P}$ on $[0, T]$ is also uniquely determined.

The maximum of two solutions is a solution. If Y^1 and Y^2 are two solutions with the same Brownian motion and initial value, then $\Delta = Y^1 - Y^2 = \int_0^\cdot (\hat{b}(Y^1_s) - \hat{b}(Y^2_s))\, ds + \int_0^\cdot \alpha(s)\, d(L^0_s(Y^1) - L^0_s(Y^2))$ has finite variation. The proof of Proposition 2.9 applies with two additional observations:

the drift terms satisfy $1_{\{Y^1>Y^2\}}\hat{b}(Y^1) + 1_{\{Y^1\leqslant Y^2\}}\hat{b}(Y^2) = \hat{b}(Y^1 \vee Y^2)$, and they do not contribute to the measures appearing in Lemma 2.8(c) and Lemma 2.7, because $\int_0^t 1_{\{Y^1_s=0\}}\, ds = 0$ by the occupation times formula and $\hat{b}(Y^1) = \hat{b}(Y^2)$ on $\{\Delta = 0\}$. Hence $Y^1 \vee Y^2$ is a solution of (2.6).

The argument of Proposition 2.10 then gives pathwise uniqueness. □

Chapter

3 SDEs Involving the Local Times of the Unknown Process

In this chapter, which follows Le Gall [8], B is a one-dimensional Brownian motion, $\phi : \mathbb{R} \to \mathbb{R}$ is a bounded Borel function and ν is a bounded signed measure on $\mathbb{R}$. We consider the stochastic differential equation

$$X_t = X_0 + \int_0^t \phi(X_s)\, dB_s + \int_{\mathbb{R}} \hat{L}_t^a(X)\, \nu(da), \tag{3.1}$$

where $\hat{L}_t^a(X)$ is the symmetric local time of the semimartingale X at level a and time t (Definition 1.21). Two cases of (3.1) are of special interest. When ν is absolutely continuous with respect to Lebesgue measure, $\nu(da) = g(a)\, da$, the occupation times formula (and the fact that $\hat{L}^a = L^a$ for all but countably many a) shows that (3.1) is the usual equation

$$X_t = X_0 + \int_0^t \phi(X_s)\, dB_s + \int_0^t (g\,\phi^2)(X_s)\, ds. \tag{3.2}$$

When $\phi = 1$ and $\nu = \beta\delta_0$, where δ_0 is the Dirac measure at 0, we get

$$X_t = X_0 + B_t + \beta \hat{L}_t^0(X), \tag{3.3}$$

whose solution is the skew Brownian motion studied in Chapter 5.

3.1 One-dimensional SDEs and local times

In this section we present results on local times and stochastic differential equations that will be used in our study of (3.1). Consider the classical Itô equation

$$dX_t = \sigma(t, X_t)\, dB_t + b(t, X_t)\, dt, \tag{3.4}$$

where B is a one-dimensional Brownian motion and $\sigma, b : \mathbb{R}_+ \times \mathbb{R} \to \mathbb{R}$ are bounded Borel functions. Our method relies on the following simple lemma [13, Lemma IX.3.3].

Lemma 3.1. *Let X be a continuous semimartingale and let $\rho :]0,\infty[\to]0,\infty[$ be a Borel function such that $\int_{0+} \frac{du}{\rho(u)} = \infty$. If, for every $t \geqslant 0$,*

$$\int_0^t \frac{1_{\{X_s>0\}}}{\rho(X_s)}\, d\langle X\rangle_s < \infty \qquad a.s.,$$

then $L_t^0(X) = 0$ for all $t \geqslant 0$, almost surely.

Proof. By the occupation times formula,

$$\int_0^t \frac{1_{\{X_s>0\}}}{\rho(X_s)}\, d\langle X\rangle_s = \int_0^\infty \frac{L_t^a(X)}{\rho(a)}\, da.$$

If $L_t^0(X) > 0$, then by right continuity $L_t^a(X) \geqslant L_t^0(X)/2$ for $a \in]0,\eta]$ and some $\eta > 0$, and the right-hand side is infinite. □

We say that σ satisfies

(A) if there exists a Borel function $\rho :]0,\infty[\to]0,\infty[$ such that $\int_{0+} \frac{du}{\rho(u)} = \infty$ and $(\sigma(t,x) - \sigma(t,y))^2 \leqslant \rho(|x-y|)$ for all t and $x \neq y$;

(B) if there exist $\varepsilon > 0$ and a bounded increasing function $f : \mathbb{R} \to \mathbb{R}$ such that $\sigma(t,x) \geqslant \varepsilon$ and $(\sigma(t,x) - \sigma(t,y))^2 \leqslant |f(x) - f(y)|$ for all t, x, y.

For instance, (A) holds if σ is Hölder continuous of order $\frac{1}{2}$ in x, uniformly in t, and (B) holds if $\sigma \geqslant \varepsilon$ and $\sigma(t,\cdot)$ has total variation bounded uniformly in t, with $f(x) = 2\|\sigma\|_\infty V(x)$ where V is a bound for the variation of $\sigma(t,\cdot)$ on $]-\infty, x]$.

Corollary 3.2. *Suppose that σ satisfies (A) or (B), and let X^1 and X^2 be two solutions of (3.4) defined on the same filtered probability space with the same Brownian motion B. Then $L_t^0(X^1 - X^2) = 0$ for all t, almost surely.*

Proof. We apply Lemma 3.1 to $X = X^1 - X^2$, for which $d\langle X\rangle_s = (\sigma(s, X_s^1) - \sigma(s, X_s^2))^2\, ds$.

Case (A). With ρ given by (A),

$$\int_0^t \frac{1_{\{X_s>0\}}}{\rho(X_s)}\, d\langle X\rangle_s = \int_0^t \frac{(\sigma(s, X_s^1) - \sigma(s, X_s^2))^2}{\rho(X_s^1 - X_s^2)}\, 1_{\{X_s^1 - X_s^2 > 0\}}\, ds \leqslant t.$$

Case (B). We take $\rho(u) = u$ and show that $\mathbb{E}\int_0^t 1_{\{X_s>0\}}\, d\langle X\rangle_s / X_s < \infty$. Let μ_f be the Stieltjes measure of the right-continuous version of f, a finite measure with $\mu_f(\mathbb{R}) \leqslant 2\|f\|_\infty$. For $x > y$ we have $f(x) - f(y) \leqslant \mu_f([y,x])$, hence, on $\{X_s > \delta\}$ with $\delta > 0$,

$$\frac{d\langle X\rangle_s}{X_s} \leqslant \frac{f(X_s^1) - f(X_s^2)}{X_s^1 - X_s^2}\, ds \leqslant \int_{\mathbb{R}} \frac{1_{\{X_s^2 \leqslant a \leqslant X_s^1\}}}{X_s^1 - X_s^2}\, \mu_f(da)\, ds.$$

For $u \in [0,1]$ let $Z^u = X^2 + u(X^1 - X^2)$. Then $Z^u = Z_0^u + \int_0^\cdot \sigma_s^u \, dB_s + \int_0^\cdot b_s^u \, ds$ with $\varepsilon \leqslant \sigma^u \leqslant K$ and $|b^u| \leqslant K$, where K bounds $|\sigma|$ and $|b|$. For $x > y$ and $\eta > 0$, $\frac{1}{x-y}\int_y^x \frac{1}{\eta} 1_{[a,a+\eta[}(z)\, dz = \int_0^1 \frac{1}{\eta} 1_{[a,a+\eta[}(y + u(x-y))\, du$, which tends to $\frac{1_{\{y \leqslant a < x\}}}{x-y}$ as $\eta \downarrow 0$. By Fatou's lemma, the bound $ds \leqslant \varepsilon^{-2} d\langle Z^u \rangle_s$ and the occupation times formula,

$$\mathbb{E}\int_0^t \frac{1_{\{X_s^2 \leqslant a < X_s^1\}}}{X_s^1 - X_s^2}\, ds \leqslant \liminf_{\eta \downarrow 0} \int_0^1 \mathbb{E}\Big[\frac{1}{\eta\varepsilon^2}\int_a^{a+\eta} L_t^x(Z^u)\, dx\Big] du \leqslant \frac{C}{\varepsilon^2},$$

where $C = \sup_{u,x} \mathbb{E} L_t^x(Z^u)$ is finite: by Tanaka's formula, $L_t^x(Z^u) \leqslant |Z_t^u - Z_0^u| + |\int_0^t \operatorname{sgn}(Z_s^u - x)\sigma_s^u \, dB_s| + Kt$, whose expectation is at most $2K\sqrt{t} + 2Kt$. The event $\{X_s^1 = a\}$ does not matter since $\int_0^t 1_{\{X_s^1 = a\}}\, ds = 0$. Integrating with respect to $\mu_f(da)$ and letting $\delta \downarrow 0$, we get $\mathbb{E}\int_0^t 1_{\{X_s > 0\}}\, d\langle X \rangle_s / X_s \leqslant 2\|f\|_\infty C/\varepsilon^2$, and Lemma 3.1 applies. □

Remark 3.3. The proof only uses the martingale parts of X^1 and X^2. The result of the corollary therefore remains valid, by localisation, if X^1 and X^2 are only assumed to satisfy $dX_t^i = \sigma(t, X_t^i)\, dB_t + dV_t^i$, where V^1 and V^2 are continuous adapted processes of finite variation.

Theorem 3.4. *Suppose that σ and b satisfy one of the following assumptions:*

(a) σ satisfies (A) and b is Lipschitz in x, uniformly in t;

(b) σ satisfies (A) and $\sigma \geqslant \varepsilon$ for some $\varepsilon > 0$;

(c) σ satisfies (B).

Then pathwise uniqueness holds for (3.4).

Remark 3.5. Each of the assumptions (a), (b) and (c) also implies weak existence for (3.4). By the theorem of Yamada and Watanabe, there is then a unique solution of (3.4) with a given initial value on any filtered probability space carrying a Brownian motion, and this solution is strong [8, 13].

Proof. Let X^1 and X^2 be two solutions of (3.4) with the same Brownian motion and the same initial value, and let $\Delta = X^1 - X^2$. By Corollary 3.2, $L^0(\Delta) = 0$.

Cases (b) and (c). Since $\sigma \geqslant \varepsilon$, uniqueness in law holds for (3.4), a result due to Stroock and Varadhan in dimension one [8]. By Tanaka's formula and $L^0(\Delta) = 0$,

$$X^1 \vee X^2 = X^2 + \Delta^+ = X^2 + \int_0^\cdot 1_{\{\Delta_s > 0\}}\, d\Delta_s,$$

so that, since $1_{\{X^1 > X^2\}}\sigma(s, X^1) + 1_{\{X^1 \leqslant X^2\}}\sigma(s, X^2) = \sigma(s, X^1 \vee X^2)$ and similarly for b, the process $X^1 \vee X^2$ is again a solution of (3.4) with the same initial value. Hence $X^1 \vee X^2$ and X^1 have the same law, and since $X^1 \vee X^2 \geqslant X^1$, the argument of Proposition 2.10 shows that $X^1 \vee X^2 = X^1$. Similarly $X^1 \vee X^2 = X^2$, so $X^1 = X^2$.

Case (a). Let K be a Lipschitz constant for b. Since $L^0(\Delta) = 0$, Tanaka's formula gives $|\Delta_t| = \int_0^t \operatorname{sgn}(\Delta_s)\, d\Delta_s$. The stochastic integral is a martingale because σ is bounded, so

$$\mathbb{E}|\Delta_t| = \mathbb{E}\int_0^t \operatorname{sgn}(\Delta_s)\Big(b(s, X_s^1) - b(s, X_s^2)\Big)\, ds \leqslant K \int_0^t \mathbb{E}|\Delta_s|\, ds,$$

and Gronwall's lemma gives $\mathbb{E}|\Delta_t| = 0$ for all t. □

Theorem 3.6 (Comparison theorem)**.** *For $i = 1, 2$, let X^i satisfy $dX_t^i = \sigma(t, X_t^i)\, dB_t + b_i(t, X_t^i)\, dt$, where σ, b_1, b_2 are bounded Borel functions. Assume that σ satisfies (A) or (B), that one of the functions b_1, b_2 is Lipschitz in x uniformly in t, that $b_1 \geqslant b_2$, and that $X_0^1 \geqslant X_0^2$ almost surely. Then, almost surely, $X_t^1 \geqslant X_t^2$ for all t.*

Proof. By Corollary 3.2 and the remark following it, the local time at 0 of $X^2 - X^1$ vanishes. Suppose that b_1 is Lipschitz with constant K. By Tanaka's formula, since $(X_0^2 - X_0^1)^+ = 0$ and the stochastic integral is a martingale,

$$\begin{aligned}\mathbb{E}[(X_t^2 - X_t^1)^+] &= \mathbb{E}\int_0^t 1_{\{X_s^2 > X_s^1\}}\Big(b_2(s, X_s^2) - b_1(s, X_s^1)\Big)\, ds \\ &\leqslant \mathbb{E}\int_0^t 1_{\{X_s^2 > X_s^1\}}\Big(b_1(s, X_s^2) - b_1(s, X_s^1)\Big)\, ds \leqslant K\int_0^t \mathbb{E}[(X_s^2 - X_s^1)^+]\, ds,\end{aligned}$$

using $b_2 \leqslant b_1$. If b_2 is Lipschitz, we use instead $b_2(s, X_s^2) - b_1(s, X_s^1) \leqslant b_2(s, X_s^2) - b_2(s, X_s^1)$. In both cases Gronwall's lemma gives $\mathbb{E}[(X_t^2 - X_t^1)^+] = 0$ for all t, and the result follows by continuity. □

We finally state a "strong" limit theorem for solutions of equations of the type (3.4), which will be used in the section on limit theorems. Its proof uses techniques similar to those of Theorems 3.4 and 3.6; we refer to Le Gall [8].

Theorem 3.7. *Let X^n $(n \geqslant 1)$ and X be continuous semimartingales on the same filtered probability space such that*

$$dX_t^n = \sigma_n(X_t^n)\, dB_t + b_n(X_t^n)\, dt, \qquad dX_t = \sigma(X_t)\, dB_t + b(X_t)\, dt,$$

where σ_n, b_n, σ and b are Borel functions on $\mathbb{R}$. Assume that

(i) σ satisfies (A) or (B);

(ii) there exist constants $\varepsilon, K > 0$ such that $\varepsilon \leqslant \sigma_n \leqslant K$ and $|b_n| \leqslant K$ for all n;

(iii) $X_0^n \to X_0$ in L^1;

(iv) $\int_{-M}^{M} |\sigma_n(u) - \sigma(u)|\, du \to 0$ and $\int_{-M}^{M} |b_n(u) - b(u)|\, du \to 0$ for every $M > 0$.

Then, for every $t \geqslant 0$,

$$\mathbb{E}\Big[\sup_{0 \leqslant s \leqslant t} |X_s^n - X_s|\Big] \longrightarrow 0.$$

3.2 The singular equation

We now turn to equations involving the local times of the unknown process. We first introduce some notation. Let $BV(\mathbb{R})$ be the space of right continuous functions $f : \mathbb{R} \to \mathbb{R}$ of bounded variation on $\mathbb{R}$ such that $f \geqslant \varepsilon$ for some $\varepsilon > 0$. For $f \in BV(\mathbb{R})$, we write $f(x-)$ for the left limit of f at x and df for the bounded signed measure associated with f, which is the derivative of f in the sense of distributions. Let $M(\mathbb{R})$ be the space of bounded signed measures ν on $\mathbb{R}$ such that $|\nu(\{x\})| < 1$ for every $x \in \mathbb{R}$. For $\nu \in M(\mathbb{R})$, ν^c denotes the continuous (diffuse) part of ν.

Lemma 3.8 (8)**.** *Let $\nu \in M(\mathbb{R})$. There exists a function f in $BV(\mathbb{R})$, unique up to a multiplicative constant, such that*

$$df(dx) + \Big(f(x) + f(x-)\Big)\nu(dx) = 0. \tag{3.5}$$

The unique such function with $f(x) \to 1$ as $x \to -\infty$ is

$$f_\nu(x) = \exp\Big(-2\nu^c(]-\infty, x])\Big) \prod_{y \leqslant x} \frac{1-\nu(\{y\})}{1+\nu(\{y\})}. \tag{3.6}$$

Proof. Since $\sum_y |\nu(\{y\})| < \infty$ and $|\nu(\{y\})| < 1$, the product converges and f_ν is bounded above and below by positive constants; it is right continuous and of bounded variation. Its diffuse part satisfies $df_\nu^c = -2f_\nu\, d\nu^c$, and at an atom y, $f_\nu(y)(1+\nu(\{y\})) = f_\nu(y-)(1-\nu(\{y\}))$, which is $f_\nu(y) - f_\nu(y-) = -(f_\nu(y) + f_\nu(y-))\nu(\{y\})$. Hence f_ν satisfies (3.5). Conversely, these two relations determine the diffuse part and the jumps of any solution f in terms of f itself, and the same computation shows that f/f_ν has no jumps and zero diffuse derivative, hence is constant. □

We are interested in the singular stochastic differential equation

$$dX_t = \phi(X_t)\, dB_t + \int_{\mathbb{R}} d_t\hat{L}_t^a(X)\, \nu(da), \tag{3.7}$$

where ϕ is in $BV(\mathbb{R})$ and ν is in $M(\mathbb{R})$. The next proposition relates (3.7) to a classical Itô equation without drift.

Proposition 3.9. *Let $\phi \in BV(\mathbb{R})$ and $\nu \in M(\mathbb{R})$, let f_ν be given by (3.6) and set $F_\nu(x) = \int_0^x f_\nu(y)\, dy$. Then X is a solution of (3.7) if and only if $Y = F_\nu(X)$ is a solution of*

$$dY_t = (\phi\, f_\nu) \circ F_\nu^{-1}(Y_t)\, dB_t. \tag{3.8}$$

Proof. Write $f = f_\nu$, $F = F_\nu$ and $f_{\mathrm{s}}(x) = \frac{1}{2}(f(x) + f(x-))$, the symmetric derivative of F. The function F is a strictly increasing bijection of $\mathbb{R}$ and a difference of convex functions, with $F'' = df$.

Let X be a solution of (3.7). Since $d\langle X\rangle = \phi^2(X)\,ds$ with $\phi \geqslant \varepsilon$, the occupation times formula shows that X spends zero Lebesgue time in the countable set of discontinuities of f, so that $f_s(X_s) = f(X_s)$ for almost every s. By the symmetric Itô–Tanaka formula (Theorem 1.23),

$$\begin{aligned} Y_t &= Y_0 + \int_0^t f_s(X_s)\,dX_s + \frac{1}{2}\int_{\mathbb{R}} \hat{L}_t^a\,df(da) \\ &= Y_0 + \int_0^t (f\phi)(X_s)\,dB_s + \int_{\mathbb{R}} \Big(\int_0^t f_s(X_s)\,d_s\hat{L}_s^a\Big)\nu(da) + \frac{1}{2}\int_{\mathbb{R}} \hat{L}_t^a\,df(da) \\ &= Y_0 + \int_0^t (f\phi)(X_s)\,dB_s + \frac{1}{2}\int_{\mathbb{R}} \hat{L}_t^a\Big(\big(f(a)+f(a-)\big)\nu(da) + df(da)\Big) \\ &= Y_0 + \int_0^t (f\phi)(X_s)\,dB_s, \end{aligned}$$

where we used that $d\hat{L}^a$ is carried by $\{X = a\}$ and (3.5). Since $X = F^{-1}(Y)$, Y satisfies (3.8).

Conversely, let Y be a solution of (3.8) and $X = G(Y)$ with $G = F^{-1}$. The function G is a difference of convex functions with $G'(y) = 1/f(G(y))$ for all but countably many y, and Y spends zero Lebesgue time in this countable set. By the symmetric Itô–Tanaka formula,

$$X_t = X_0 + \int_0^t \frac{(\phi f)(X_s)}{f(X_s)}\,dB_s + V_t = X_0 + \int_0^t \phi(X_s)\,dB_s + V_t, \qquad V_t = \frac{1}{2}\int_{\mathbb{R}} \hat{L}_t^{F(a)}(Y)\,d\Big(\frac{1}{f}\Big)(a),$$

where we changed variables $b = F(a)$ in the measure $G''(db)$. Since $d\langle Y\rangle = f(X)^2\,d\langle X\rangle$, the occupation times formula and right continuity give $L^{F(a)}(Y) = f(a)L^a(X)$ and $L^{F(a)-}(Y) = f(a-)L^{a-}(X)$ for every a, so that

$$V_t = \frac{1}{4}\int_{\mathbb{R}} \Big(f(a)L_t^a(X) + f(a-)L_t^{a-}(X)\Big)\,d\Big(\frac{1}{f}\Big)(a).$$

The diffuse part of $d(1/f)$ is $-df^c/f^2 = 2\nu^c/f$, and since $L^a = L^{a-} = \hat{L}^a$ for all but countably many a, it contributes $\int \hat{L}_t^a(X)\,\nu^c(da)$. Let a be an atom of ν, $c = \nu(\{a\})$ and $r = f(a)/f(a-) = (1-c)/(1+c)$. The contribution of a to V is $\frac{1}{4}\Big(L^a(X)(1-r) + L^{a-}(X)(r^{-1}-1)\Big)$, and this is also $\int 1_{\{X_s=a\}}\,dV_s$. By the jump formula of Theorem 1.16, $L^a(X) - L^{a-}(X)$ equals twice this quantity, which forces $L^{a-}(X) = r\,L^a(X)$. Hence the contribution of a is $\frac{1}{2}(1-r)L^a(X) = \frac{c}{1+c}L^a(X) = c\,\hat{L}^a(X)$, since $\hat{L}^a(X) = \frac{1+r}{2}L^a(X) = \frac{1}{1+c}L^a(X)$. Therefore $V_t = \int_{\mathbb{R}} \hat{L}_t^a(X)\,\nu(da)$, and X solves (3.7). □

The function $(\phi f_\nu)\circ F_\nu^{-1}$ belongs to $BV(\mathbb{R})$, since F_ν^{-1} is an increasing bijection. We have thus shown that (3.7) is equivalent to an equation of the same type with $\nu = 0$.

Theorem 3.10. *Let $\phi \in BV(\mathbb{R})$ and $\nu \in M(\mathbb{R})$. Then existence and pathwise uniqueness hold for (3.7).*

Proof. By Proposition 3.9, it suffices to consider (3.8), whose coefficient $h = (\phi f_\nu)\circ F_\nu^{-1}$ is in $BV(\mathbb{R})$. If $V_h(x)$ denotes the total variation of h on $]-\infty, x]$, then $(h(x)-h(y))^2 \leqslant$

$2\|h\|_\infty |V_h(x) - V_h(y)|$, so h satisfies (B), and pathwise uniqueness follows from Theorem 3.4 (c). Weak existence for (3.8) holds since h is bounded and bounded below by a positive constant (for instance by a time change), and existence of a strong solution then follows from the Yamada–Watanabe theorem. □

3.3 Limit theorems

In this section we use Theorem 3.7 to prove limit theorems for solutions of equations of the form (3.7). They show that any solution of (3.7) is the limit, in a strong sense, of solutions of classical Itô equations.

Theorem 3.11. *Let $\nu_n \in M(\mathbb{R})$ and $\phi_n \in BV(\mathbb{R})$ $(n \geqslant 1)$, and suppose that there exist constants $\varepsilon, M > 0$ such that, for all n and x,*

$$|\nu_n|(\mathbb{R}) \leqslant M, \qquad \varepsilon \leqslant \phi_n(x) \leqslant M, \qquad |\nu_n(\{x\})| \leqslant 1 - \varepsilon.$$

Let $(\Omega, \mathcal{F}, (\mathcal{F}_t), \mathbb{P})$ be a filtered probability space carrying a Brownian motion B, and for $n \geqslant 1$ let X^n be the solution of

$$X_t^n = X_0^n + \int_0^t \phi_n(X_s^n)\, dB_s + \int_{\mathbb{R}} \hat{L}_t^a(X^n)\, \nu_n(da),$$

where $X_0^n \to X_0$ in L^1. Assume further that there exist two functions ϕ, f in $BV(\mathbb{R})$ such that, for every $K > 0$,

$$\int_{-K}^{K} |\phi_n - \phi|(x)\, dx \longrightarrow 0, \qquad \int_{-K}^{K} |f_{\nu_n} - f|(x)\, dx \longrightarrow 0,$$

and set $\nu(dx) = -\dfrac{df(dx)}{f(x) + f(x-)}$. Then $\nu \in M(\mathbb{R})$, and for every $t \geqslant 0$,

$$\mathbb{E}\Big[\sup_{0 \leqslant s \leqslant t} |X_s^n - X_s|\Big] \longrightarrow 0,$$

where X is the solution of

$$X_t = X_0 + \int_0^t \phi(X_s)\, dB_s + \int_{\mathbb{R}} \hat{L}_t^a(X)\, \nu(da).$$

Remark 3.12. The limit of the sequence (ν_n), if it exists, need not coincide with ν. For instance, let $X_0^n = 0$, $\phi_n = 1$ and $\nu_n(dx) = g_n(x)\, dx$, where $g_n \geqslant 0$, $\int_{\mathbb{R}} g_n(x)\, dx = A > 0$ and g_n vanishes outside $[-\varepsilon_n, \varepsilon_n]$ with $\varepsilon_n \to 0$. Then $\nu_n \to A\delta_0$ weakly, while $f_{\nu_n}(x) = \exp(-2\int_{-\infty}^{x} g_n)$ converges to $f = 1_{]-\infty,0[} + e^{-2A} 1_{[0,\infty[}$. The theorem shows that X^n converges to the solution of

$$X_t = B_t + \frac{1 - e^{-2A}}{1 + e^{-2A}}\, \hat{L}_t^0(X) = B_t + \tanh(A)\, \hat{L}_t^0(X),$$

and not to the solution of $X_t = B_t + A\hat{L}_t^0(X)$, which does not even exist when $A > 1$.

Proof of Theorem 3.11. Since $\inf f > 0$ and df is bounded, ν is a bounded signed measure, and $|\nu(\{x\})| = |f(x) - f(x-)|/(f(x) + f(x-)) < 1$, so $\nu \in M(\mathbb{R})$; moreover f solves (3.5) for this ν, so $f = c\, f_\nu$ for a constant $c > 0$, and Proposition 3.9 holds with f in place of f_ν. Let

$$f_n = f_{\nu_n}, \qquad F_n(x) = \int_0^x f_n(y)\, dy, \qquad h_n = (\phi_n f_n) \circ F_n^{-1},$$

and define F and $h = (\phi f) \circ F^{-1}$ in the same way. Since $|\log \frac{1-c}{1+c}| \leqslant C_\varepsilon |c|$ for $|c| \leqslant 1 - \varepsilon$, the bounds on ν_n give constants $0 < \kappa \leqslant K'$ with $\kappa \leqslant f_n \leqslant K'$ for all n; hence there are constants $\varepsilon', M' > 0$ with $\varepsilon' \leqslant h_n \leqslant M'$ for all n. From the L^1_{loc} convergence of f_n and ϕ_n, $F_n \to F$ and $F_n^{-1} \to F^{-1}$ locally uniformly, and one checks that $\int_{-K}^{K} |h_n - h|(y)\, dy \to 0$ for every $K > 0$.

By Proposition 3.9, $Y^n = F_n(X^n)$ satisfies $Y_t^n = F_n(X_0^n) + \int_0^t h_n(Y_s^n)\, dB_s$. Let Y be the solution of $Y_t = F(X_0) + \int_0^t h(Y_s)\, dB_s$; since $h \in BV(\mathbb{R})$, it satisfies (B), as in the proof of Theorem 3.10. Moreover

$$\begin{aligned}\mathbb{E}|F_n(X_0^n) - F(X_0)| &\leqslant \mathbb{E}|F_n(X_0^n) - F_n(X_0)| + \mathbb{E}|F_n(X_0) - F(X_0)| \\ &\leqslant K'\, \mathbb{E}|X_0^n - X_0| + \mathbb{E}|F_n(X_0) - F(X_0)| \longrightarrow 0,\end{aligned}$$

by dominated convergence for the last term, since $|F_n(x) - F(x)| \leqslant 2K'|x|$. Theorem 3.7 therefore gives $\mathbb{E}[\sup_{s \leqslant t} |Y_s^n - Y_s|] \to 0$. Since $|F_n^{-1}(y) - F_n^{-1}(y')| \leqslant |y - y'|/\kappa$ and $F_n^{-1} \to F^{-1}$ locally uniformly, it follows that

$$\mathbb{E}\Big[\sup_{s \leqslant t} |X_s^n - F^{-1}(Y_s)|\Big] \longrightarrow 0.$$

Finally, by Proposition 3.9, $X = F^{-1}(Y)$ solves the equation with coefficients ϕ and ν. □

Corollary 3.13. *Let $\nu \in M(\mathbb{R})$, let $X_0 \in L^1$, and let X be the solution of*

$$X_t = X_0 + B_t + \int_{\mathbb{R}} \hat{L}_t^a(X)\, \nu(da).$$

For $n \geqslant 1$, set $X_t^n = \frac{1}{n} X_{n^2 t}$. Then X^n converges in law to the skew Brownian motion started at 0 with parameter β given by

$$\frac{1-\beta}{1+\beta} = \exp\Big(-2\nu^c(\mathbb{R})\Big) \prod_{y \in \mathbb{R}} \frac{1 - \nu(\{y\})}{1 + \nu(\{y\})} = \lim_{x \to +\infty} f_\nu(x).$$

Remark 3.14. Recall that the skew Brownian motion with parameter $\beta \in [-1, 1]$ is the unique solution of $X_t = B_t + \beta \hat{L}_t^0(X)$. In the case $\nu(da) = g(a)\, da$, studied by Rosenkrantz [14], the corollary gives $\frac{1-\beta}{1+\beta} = \exp\big(-2 \int_{\mathbb{R}} g(x)\, dx\big)$.

Proof. For each n, let ν_n be the image of ν under $a \mapsto a/n$, that is $\nu_n(A) = \nu(nA)$. The process $W_t^n = \frac{1}{n} B_{n^2 t}$ is a Brownian motion, and the occupation times formula gives $L_t^b(X^n) = \frac{1}{n} L_{n^2 t}^{nb}(X)$, and similarly for symmetric local times. Hence

$$X_t^n = \frac{1}{n} X_0 + W_t^n + \int_{\mathbb{R}} \hat{L}_t^b(X^n)\, \nu_n(db),$$

so that, by uniqueness in law, X^n has the same law as the solution Y^n of

$$Y_t^n = \frac{1}{n}X_0 + B_t + \int_{\mathbb{R}} \hat{L}_t^b(Y^n)\,\nu_n(db).$$

The measures ν_n satisfy the assumptions of Theorem 3.11 with $\phi_n = 1$, and $f_{\nu_n}(x) = f_\nu(nx)$ converges boundedly to 1 for $x < 0$ and to $f_\nu(+\infty)$ for $x > 0$. The limiting function $f = 1_{]-\infty,0[} + f_\nu(+\infty)1_{[0,\infty[}$ corresponds to $\nu = \beta\delta_0$ with $\frac{1-\beta}{1+\beta} = f_\nu(+\infty)$, and Theorem 3.11 shows that Y^n converges in L^1, uniformly on compact time intervals, to the skew Brownian motion of parameter β started at 0. □

Remark 3.15. Corollary 3.13 gives convergence to a skew Brownian motion with parameter $\beta \in]-1, 1[$. The extreme values $\beta = \pm 1$, which correspond to reflected Brownian motion, cannot be obtained in this way. One can however prove a similar convergence result towards reflected Brownian motion: if $g_n \in L^1(\mathbb{R})$, $g_n \geqslant 0$, g_n vanishes outside $[-\varepsilon_n, \varepsilon_n]$ with $\varepsilon_n \to 0$ and $\int_{\mathbb{R}} g_n(x)\,dx \to \infty$, then the solutions of $X_t^n = B_t + \int_0^t g_n(X_s^n)\,ds$ converge in law to $|B|$.

As a last application of Theorem 3.11, we prove a comparison theorem.

Theorem 3.16. *Let $\nu, \mu \in M(\mathbb{R})$ and $\phi \in BV(\mathbb{R})$. Let X and Y be the solutions, with respect to the same Brownian motion, of*

$$X_t = X_0 + \int_0^t \phi(X_s)\,dB_s + \int_{\mathbb{R}} \hat{L}_t^a(X)\,\nu(da), \qquad Y_t = Y_0 + \int_0^t \phi(Y_s)\,dB_s + \int_{\mathbb{R}} \hat{L}_t^a(Y)\,\mu(da),$$

where $X_0, Y_0 \in L^1$. If $X_0 \geqslant Y_0$ and $\nu \geqslant \mu$, then almost surely $X_t \geqslant Y_t$ for all t.

Proof. Suppose first that $\nu(da) = g(a)\,da$ and $\mu(da) = k(a)\,da$, where $g \geqslant k$ are Lipschitz, and that ϕ is Lipschitz. By (3.2), X and Y solve classical Itô equations with drifts $g\phi^2 \geqslant k\phi^2$, which are Lipschitz, and ϕ satisfies (A). Theorem 3.6 then gives $X \geqslant Y$.

In the general case, we approximate. Let $\tilde{\nu} = \nu^c + \sum_a \operatorname{artanh}(\nu(\{a\}))\,\delta_a$ and define $\tilde{\mu}$ similarly; since $\nu - \mu \geqslant 0$, both its diffuse part $\nu^c - \mu^c$ and its atoms are nonnegative, so $\tilde{\nu} \geqslant \tilde{\mu}$. Let $\nu_n = \tilde{\nu} * \rho_n$ and $\mu_n = \tilde{\mu} * \rho_n$, where (ρ_n) is a smooth approximation of the identity, and let $\phi_n = \phi * \rho_n$. Then $\nu_n \geqslant \mu_n$ have Lipschitz densities, ϕ_n is Lipschitz with $\inf \phi \leqslant \phi_n \leqslant \sup \phi$, and $\phi_n \to \phi$ in L^1_{loc}. Moreover $f_{\nu_n}(x) = \exp(-2\nu_n(]-\infty, x]))$ converges, at every x which is not an atom of ν, to $\exp(-2\tilde{\nu}(]-\infty, x])) = f_\nu(x)$, because $\exp(-2\operatorname{artanh} c) = \frac{1-c}{1+c}$; as in Remark 3.12, an atom c must be spread with mass $\operatorname{artanh}(c)$, not c. Hence $f_{\nu_n} \to f_\nu$ and similarly $f_{\mu_n} \to f_\mu$ in L^1_{loc}, and the measures ν_n, μ_n have total mass at most $|\tilde{\nu}|(\mathbb{R})$ and $|\tilde{\mu}|(\mathbb{R})$ and no atoms. Let X^n and Y^n be the solutions associated with (ϕ_n, ν_n) and (ϕ_n, μ_n) and the initial values X_0, Y_0. By the first step, $X^n \geqslant Y^n$, and by Theorem 3.11, $X_t^n \to X_t$ and $Y_t^n \to Y_t$ in L^1 for every t. Hence $X_t \geqslant Y_t$ almost surely for every t, and the result follows by continuity. □

Chapter

4 Parabolic Partial Differential Equations

In this chapter we show how Brownian motion can be used to construct and characterise solutions of parabolic equations. We follow the presentation of Durrett [1]. The chapter serves as a model for Chapter 5: there, Brownian motion is replaced by the skew Brownian motion, and the same martingale arguments produce a heat equation with a *transmission condition* at 0 (see Section 4.2).

4.1 Parabolic equations

Let B be a d-dimensional Brownian motion, and write $\mathbb{E}_x$ for the expectation when $B_0 = x$. We consider the equations

$$\frac{\partial u}{\partial t} = \frac{1}{2}\Delta u, \tag{4.1}$$

$$\frac{\partial u}{\partial t} = \frac{1}{2}\Delta u + g, \tag{4.2}$$

$$\frac{\partial u}{\partial t} = \frac{1}{2}\Delta u + c\,u, \tag{4.3}$$

in $(0,\infty)\times\mathbb{R}^d$, subject to the boundary condition

$$u \text{ is continuous at each point of } \{0\}\times\mathbb{R}^d \text{ and } u(0,x) = f(x) \text{ for } x \in \mathbb{R}^d. \tag{4.4}$$

Here

$$\Delta u = \frac{\partial^2 u}{\partial x_1^2} + \cdots + \frac{\partial^2 u}{\partial x_d^2}$$

is the Laplacian in the space variables. By a *classical solution* we mean one that has enough derivatives for the equation to make sense, that is, $u \in C^{1,2}$: the functions on $(0,\infty)\times\mathbb{R}^d$ that

have one continuous derivative with respect to t and two continuous derivatives with respect to the x_i. The continuity of u in (4.4) is needed to connect the equation, which holds in $(0,\infty)\times\mathbb{R}^d$, with the initial value $u(0,x)=f(x)$, which holds on $\{0\}\times\mathbb{R}^d$. Note that (4.4) cannot hold unless $f:\mathbb{R}^d\to\mathbb{R}$ is continuous.

We will see that, under suitable assumptions, the solutions of (4.1), (4.2) and (4.3) are given respectively by

$$\mathbb{E}_x f(B_t), \qquad \mathbb{E}_x\Big(f(B_t)+\int_0^t g(t-s,B_s)\,ds\Big), \qquad \mathbb{E}_x\Big(f(B_t)\exp\Big(\int_0^t c(t-s,B_s)\,ds\Big)\Big).$$

The three cases are treated by the same procedure, in four steps: find a local martingale, deduce uniqueness, show that the candidate is a solution when it is smooth, and give conditions for smoothness. We carry it out for the heat equation (4.1) and for the Feynman–Kac equation (4.3); the inhomogeneous equation (4.2) is handled in the same way (see Durrett [1]).

4.1.1 The heat equation

Bounded initial data

In this section we consider the problem

(A) $u\in C^{1,2}$ and $\dfrac{\partial u}{\partial t}=\dfrac{1}{2}\Delta u$ in $(0,\infty)\times\mathbb{R}^d$;

(B) u is continuous at each point of $\{0\}\times\mathbb{R}^d$ and $u(0,x)=f(x)$.

The equation takes its name from the fact that, if the units are chosen suitably, $u(t,x)$ is the temperature at the point $x\in\mathbb{R}^d$ at time t when the temperature profile at time 0 is f.

The first step is to find a local martingale.

Theorem 4.1. *If u satisfies (A), then for every $t>0$ the process $M_s=u(t-s,B_s)$ is a local martingale on $[0,t)$.*

Proof. Apply Itô's formula to $u(x_0,x_1,\dots,x_d)$ with $X_s^0=t-s$ and $X_s^i=B_s^i$ for $1\leqslant i\leqslant d$. Since X^0 has bounded variation with $dX_r^0=-dr$, and the X^i, $1\leqslant i\leqslant d$, are independent Brownian motions,

$$\langle X^i,X^j\rangle_r=\begin{cases} r & \text{if } 1\leqslant i=j\leqslant d,\\ 0 & \text{otherwise,}\end{cases}$$

and for $s<t$

$$u(t-s,B_s)-u(t,B_0)=\int_0^s\Big(-\frac{\partial u}{\partial t}+\frac{1}{2}\Delta u\Big)(t-r,B_r)\,dr+\int_0^s\nabla u(t-r,B_r)\cdot dB_r.$$

By (A) the first integral vanishes, and the second is a local martingale. □

The next step is a uniqueness theorem. We write u for a generic solution and

$$v(t,x) = \mathbb{E}_x f(B_t)$$

for our candidate solution.

Theorem 4.2. *If u is a bounded solution of (A)–(B), then $u = v$.*

Proof. Fix $t > 0$. Since u is bounded, the local martingale $M_s = u(t-s, B_s)$, $0 \leqslant s < t$, of Theorem 4.1 is a bounded martingale. By the martingale convergence theorem, $M_t = \lim_{s \uparrow t} M_s$ exists almost surely. As $s \uparrow t$, $(t-s, B_s) \to (0, B_t)$, so (B) gives $M_t = f(B_t)$. Since M is uniformly integrable,

$$u(t,x) = \mathbb{E}_x M_0 = \mathbb{E}_x M_t = \mathbb{E}_x f(B_t) = v(t,x). \qquad \square$$

Theorem 4.2 tells us what a bounded solution must be. The next step is to find conditions under which v is a solution. We first show that if v is smooth enough, then it satisfies the equation.

Theorem 4.3. *Suppose that f is bounded. If $v \in C^{1,2}$, then v satisfies (A).*

Proof. Fix $t > 0$. The Markov property implies that, for $0 \leqslant s < t$,

$$\mathbb{E}_x\Big(f(B_t) \,\Big|\, \mathcal{F}_s\Big) = \mathbb{E}_{B_s} f(B_{t-s}) = v(t-s, B_s),$$

so $v(t-s, B_s)$ is a martingale on $[0,t)$. If $v \in C^{1,2}$, the computation in the proof of Theorem 4.1 gives

$$v(t-s, B_s) - v(t, B_0) = \int_0^s \Big(-\frac{\partial v}{\partial t} + \frac{1}{2}\Delta v\Big)(t-r, B_r)\, dr + \text{a local martingale.}$$

The left-hand side is a local martingale, so the integral on the right is a continuous local martingale of locally bounded variation, starting from 0; hence it is identically 0 almost surely.

Let $\varphi = -\partial_t v + \frac{1}{2}\Delta v$, which is continuous. Suppose that $\varphi(t_0, x_0) \neq 0$, say $\varphi > 0$ on the ball of radius $\delta \in (0, t_0)$ centred at (t_0, x_0). Take $t = t_0$ and $B_0 = x_0$, and let $\tau = \inf\{r : |B_r - x_0| \geqslant \delta/2\} \wedge \delta/2 > 0$. For $r < \tau$, the point $(t_0 - r, B_r)$ lies in that ball, so $\int_0^s \varphi(t_0 - r, B_r)\, dr > 0$ for $0 < s \leqslant \tau$, which is a contradiction. Hence $\varphi \equiv 0$. $\square$

It is easy to give conditions under which v satisfies the boundary condition (B).

Theorem 4.4. *If f is bounded and continuous, then v satisfies (B).*

Proof. Under $\mathbb{P}_x$, $B_t - B_0$ has the law of $t^{1/2}N$, where N is a standard normal vector. So if $t_n \to 0$ and $x_n \to x$, the bounded convergence theorem gives

$$v(t_n, x_n) = \mathbb{E} f\Big(x_n + t_n^{1/2} N\Big) \longrightarrow f(x). \qquad \square$$

The final step in showing that v is a solution is to find conditions that guarantee that it is smooth. In this case the computations are not difficult.

Theorem 4.5. *If f is bounded, then $v \in C^{1,2}$, and hence v satisfies (A).*

Proof. By definition,

$$v(t,x) = \mathbb{E}_x f(B_t) = \int p_t(x,y)\, f(y)\, dy, \qquad p_t(x,y) = (2\pi t)^{-d/2} \exp\Big(-\frac{|x-y|^2}{2t}\Big).$$

A little calculus gives

$$\begin{aligned}
\frac{\partial}{\partial x_i} p_t(x,y) &= -\frac{x_i - y_i}{t}\, p_t(x,y), \\
\frac{\partial^2}{\partial x_i^2} p_t(x,y) &= \frac{(x_i-y_i)^2 - t}{t^2}\, p_t(x,y), \\
\frac{\partial^2}{\partial x_i\, \partial x_j} p_t(x,y) &= \frac{(x_i-y_i)(x_j-y_j)}{t^2}\, p_t(x,y), \qquad i \neq j, \\
\frac{\partial}{\partial t} p_t(x,y) &= \Big(-\frac{d}{2t} + \frac{|x-y|^2}{2t^2}\Big)\, p_t(x,y).
\end{aligned}$$

Fix $0 < a < b < \infty$ and $x_0 \in \mathbb{R}^d$. For $t \in [a,b]$ and $|x - x_0| \leqslant 1$, each of these derivatives is bounded in absolute value by

$$h(y) = C\left(1 + |y - x_0|^2\right) \exp\Big(-\frac{\big((|y-x_0|-1)^+\big)^2}{2b}\Big),$$

for a constant C depending only on a, b and d, and h is integrable. Since f is bounded, $h\,\|f\|_\infty$ dominates all the integrands, so the dominated convergence theorem shows that the integrals $\int \partial p_t(x,y)\, f(y)\, dy$ of each of the derivatives above are finite and continuous in $(t,x) \in (0,\infty)\times\mathbb{R}^d$. The domination also gives condition (iii) of Lemma 4.6 below. Applying that lemma once with $K = p$, and once more with $K = \partial p/\partial x_i$ (and in the variable t in the same way), shows that the derivatives of v can be computed by differentiating under the integral sign. Hence $v \in C^{1,2}$, and Theorem 4.3 shows that v satisfies (A). □

The following lemma on differentiating under the integral sign is lengthy to state but short to prove, since we assume everything the proof needs.

Lemma 4.6. *Let $(S, \mathcal{S}, m)$ be a σ-finite measure space and $g : S \to \mathbb{R}$ be measurable. Let $G \subset \mathbb{R}^d$ be open, $x^* \in G$, $1 \leqslant i \leqslant d$, and $h_0 > 0$ such that $x^* + \theta e_i \in G$ for $|\theta| \leqslant h_0$. Suppose that:*

(i) $K : G \times S \to \mathbb{R}$ is measurable, $\int_S |K(x,y)g(y)|\, dm(y) < \infty$ for $x \in G$, and $u(x) = \int_S K(x,y)\, g(y)\, dm(y)$;

(ii) *there is a measurable* $\partial_i K : G \times S \to \mathbb{R}$ *such that* $K(x^* + he_i, y) - K(x^*, y) = \int_0^h \partial_i K(x^* + \theta e_i, y)\, d\theta$ *for* $|h| \leqslant h_0$ *and* $y \in S$*;*

(iii) $\int_S \int_{-h_0}^{h_0} |\partial_i K(x^* + \theta e_i, y)\, g(y)|\, d\theta\, dm(y) < \infty$*;*

(iv) *the function* $w(x) = \int_S \partial_i K(x, y)\, g(y)\, dm(y)$ *is well defined near* x^* *and continuous at* x^**.*

Then $\partial u/\partial x_i$ *exists at* x^* *and* $\frac{\partial u}{\partial x_i}(x^*) = w(x^*)$*.*

Proof. By (i), (ii) and Fubini's theorem, which is justified by (iii), for $0 < |h| \leqslant h_0$,

$$\begin{aligned} u(x^* + he_i) - u(x^*) &= \int_S \Big(K(x^* + he_i, y) - K(x^*, y)\Big) g(y)\, dm(y) \\ &= \int_0^h \int_S \partial_i K(x^* + \theta e_i, y)\, g(y)\, dm(y)\, d\theta = \int_0^h w(x^* + \theta e_i)\, d\theta. \end{aligned}$$

Dividing by h and letting $h \to 0$, the result follows from (iv). □

Unbounded initial data

For some applications the assumption that f is bounded is too restrictive. To see which unbounded f we can allow, observe that at the bare minimum we need $\mathbb{E}_x|f(B_t)| < \infty$ for all t and x. Since

$$\mathbb{E}_x|f(B_t)| = \int (2\pi t)^{-d/2} \exp\Big(-\frac{|x-y|^2}{2t}\Big) |f(y)|\, dy,$$

a condition that guarantees this for locally bounded f is

$$|x|^{-2} \log^+ |f(x)| \longrightarrow 0 \qquad \text{as } |x| \to \infty. \tag{4.5}$$

Indeed, (4.5) means that for every $\varepsilon > 0$ there is C_ε with $|f(x)| \leqslant C_\varepsilon e^{\varepsilon |x|^2}$. Replacing the bounded convergence theorem by the dominated convergence theorem in the proofs of Theorems 4.4 and 4.5, it is not hard to show the following.

Theorem 4.7. *If* f *is continuous and satisfies* (4.5)*, then* $v(t,x) = \mathbb{E}_x f(B_t)$ *satisfies (A) and (B).*

Remark 4.8. The uniqueness argument of Theorem 4.2 uses the boundedness of u in an essential way, and uniqueness does fail without a growth restriction: a classical example of Tychonoff gives a solution of (A)–(B) with $f \equiv 0$ that is not identically zero. Uniqueness holds again in classes of functions with controlled growth, such as $|u(t,x)| \leqslant Ce^{a|x|^2}$ on $[0,T] \times \mathbb{R}^d$; see Friedman [3].

4.1.2 The Feynman–Kac formula

In this section we add cu to the right-hand side of the heat equation. That is, we study

(A) $u \in C^{1,2}$ and $\dfrac{\partial u}{\partial t} = \dfrac{1}{2}\Delta u + c\,u$ in $(0,\infty)\times\mathbb{R}^d$;

(B) u is continuous at each point of $\{0\}\times\mathbb{R}^d$ and $u(0,x) = f(x)$.

If $c(t,x) \leqslant 0$, this equation describes heat flow with cooling: $u(t,x)$ is the temperature at the point x at time t when heat at x at time t dissipates at the rate $-c(t,x)$.

As before, the first step is to find a local martingale.

Theorem 4.9. *Fix $t > 0$ and let $c_s^t = \int_0^s c(t-r, B_r)\,dr$. If u satisfies (A), then*

$$M_s = u(t-s, B_s)\exp(c_s^t)$$

is a local martingale on $[0,t)$.

Proof. Apply Itô's formula with $X_s^0 = t-s$, $X_s^i = B_s^i$ for $1 \leqslant i \leqslant d$, and $X_s^{d+1} = c_s^t$. The processes X^0 and X^{d+1} have bounded variation, and the brackets of the X^i are as in the proof of Theorem 4.1, so

$$\begin{aligned} u(t-s,B_s)\exp(c_s^t) - u(t,B_0) = &\int_0^s -\frac{\partial u}{\partial t}(t-r,B_r)\exp(c_r^t)\,dr + \int_0^s \exp(c_r^t)\nabla u(t-r,B_r)\cdot dB_r \\ &+ \int_0^s u(t-r,B_r)\exp(c_r^t)\,dc_r^t + \frac{1}{2}\int_0^s \Delta u(t-r,B_r)\exp(c_r^t)\,dr. \end{aligned}$$

Using $dc_r^t = c(t-r,B_r)\,dr$ and rearranging,

$$\begin{aligned} u(t-s,B_s)\exp(c_s^t) - u(t,B_0) = &\int_0^s \Big(-\frac{\partial u}{\partial t} + c\,u + \frac{1}{2}\Delta u\Big)(t-r,B_r)\exp(c_r^t)\,dr \\ &+ \int_0^s \exp(c_r^t)\nabla u(t-r,B_r)\cdot dB_r. \end{aligned}$$

By (A) the first integral vanishes, and the second is a local martingale. □

The next step, again, is a uniqueness result. Our candidate solution is now

$$v(t,x) = \mathbb{E}_x\Big(f(B_t)\exp(c_t^t)\Big).$$

Theorem 4.10. *Suppose that c is bounded. If u is a solution of (A)–(B) that is bounded on $[0,T]\times\mathbb{R}^d$ for every $T < \infty$, then $u = v$.*

Proof. Fix $t > 0$. If $|c| \leqslant M$ and $|u| \leqslant K$ on $[0,t]\times\mathbb{R}^d$, then $|M_s| \leqslant Ke^{Mt}$ for $0 \leqslant s < t$, so the local martingale of Theorem 4.9 is a bounded martingale. By (B) and the continuity of $s \mapsto c_s^t$,

$$M_t = \lim_{s\uparrow t} M_s = f(B_t)\exp(c_t^t),$$

and since M is uniformly integrable, $u(t,x) = \mathbb{E}_x M_0 = \mathbb{E}_x M_t = v(t,x)$. □

As before, it is easy to show that if v is smooth enough, it is a solution.

Theorem 4.11. *Suppose that f is bounded and that c is bounded and continuous. If $v \in C^{1,2}$, then v satisfies (A).*

Proof. Fix $t > 0$ and $0 \leqslant s < t$. Since $c_t^t - c_s^t = \int_0^{t-s} c\Big((t-s)-r, B_{s+r}\Big)\, dr$, the Markov property implies that

$$\mathbb{E}_x\Big(f(B_t)\exp(c_t^t)\,\Big|\,\mathcal{F}_s\Big) = \exp(c_s^t)\,\mathbb{E}_{B_s}\Big(f(B_{t-s})\exp(c_{t-s}^{t-s})\Big) = \exp(c_s^t)\,v(t-s,B_s).$$

The left-hand side is a martingale, so the right-hand side is also. If $v \in C^{1,2}$, the computation in the proof of Theorem 4.9 shows that

$$v(t-s,B_s)\exp(c_s^t) - v(t,B_0) = \int_0^s \Big(-\frac{\partial v}{\partial t} + c\,v + \frac{1}{2}\Delta v\Big)(t-r,B_r)\exp(c_r^t)\,dr + \text{a local martingale.}$$

As in the proof of Theorem 4.3, the integral is a continuous local martingale of locally bounded variation, hence identically 0. Since $\exp(c_r^t) > 0$ and $-\partial_t v + c\,v + \frac{1}{2}\Delta v$ is continuous, the same argument shows that $-\partial_t v + c\,v + \frac{1}{2}\Delta v \equiv 0$. □

The next step is to give conditions guaranteeing that v satisfies (B). As before, we begin with the case in which everything is bounded.

Theorem 4.12. *If c is bounded and f is bounded and continuous, then v satisfies (B).*

Proof. If $|c| \leqslant M$, then $e^{-Mt} \leqslant \exp(c_t^t) \leqslant e^{Mt}$, so $|\exp(c_t^t) - 1| \leqslant e^{Mt} - 1$. Hence

$$\Big|v(t,x) - \mathbb{E}_x f(B_t)\Big| \leqslant \|f\|_\infty\, \mathbb{E}_x\Big|\exp(c_t^t) - 1\Big| \leqslant \|f\|_\infty\Big(e^{Mt} - 1\Big) \longrightarrow 0$$

as $t \to 0$, uniformly in x. By Theorem 4.4, $(t,x) \mapsto \mathbb{E}_x f(B_t)$ is continuous at each point of $\{0\} \times \mathbb{R}^d$, and the result follows. □

This brings us to the problem of determining when v is smooth enough to be a solution. Here the computations are more involved than for the heat equation, and we only state the result; see Durrett [1] and Friedman [3].

Theorem 4.13. *Suppose that f is bounded and Hölder continuous, and that c is bounded and locally Hölder continuous. Then $v \in C^{1,2}$, and hence v satisfies (A).*

4.2 From the heat equation to the transmission problem

The arguments above use only two properties of Brownian motion: the Markov property, and Itô's formula for $u(t-s, B_s)$. In Chapter 5 we replace B by the skew Brownian motion

$$X_t = x + W_t + \beta \hat{L}_t^0(X), \qquad \beta \in [-1,1],$$

where $\hat{L}^0(X)$ is the symmetric local time of X at 0 (Definition 1.21). Away from 0, X behaves like a Brownian motion, so $u(t,x) = \mathbb{E}_x f(X_t)$ again solves the heat equation (4.1) on $(0,\infty)\times(\mathbb{R}\setminus\{0\})$. At 0, Itô's formula must be replaced by the Itô–Tanaka formula (Theorem 1.23), which produces an additional local time term. Requiring that this term vanish, as the dr-integral vanished in Theorem 4.3, gives the *transmission condition*

$$(1+\beta)\frac{\partial u}{\partial x}(t,0^+) = (1-\beta)\frac{\partial u}{\partial x}(t,0^-),$$

together with the continuity of $u(t,\cdot)$ at 0. This is carried out in Section 5.2.

Chapter

5 Skew Brownian Motion

Let $\beta \in [-1, 1]$. A *skew Brownian motion* of parameter β started at x is a solution of

$$X_t = x + W_t + \beta \hat{L}_t^0(X), \tag{5.1}$$

where W is a Brownian motion and $\hat{L}^0(X)$ is the symmetric local time of X at 0 (Definition 1.21). This process was introduced by Itô and McKean [5] and studied by Walsh [16] and Harrison and Shepp [4]; a detailed survey is Lejay [9]. For $\beta = 0$ it is a Brownian motion, and for $\beta = \pm 1$ it is a reflected Brownian motion.

When $|\beta| < 1$, (5.1) is the equation (3.7) of Chapter 3 with $\phi \equiv 1$ and $\nu = \beta\delta_0 \in M(\mathbb{R})$, so by Theorem 3.10 it has a unique strong solution, and pathwise uniqueness holds; the same is true for $|\beta| \leqslant 1$ by Harrison and Shepp [4]. It is also the equation of Remark 2.2 with a constant coefficient. In particular X is a strong Markov process. Throughout this chapter, $\mathbb{P}_x$ and $\mathbb{E}_x$ refer to the solution started at x, and we write $\hat{L}_t = \hat{L}_t^0(X)$.

Remark 5.1. For $\nu = \beta\delta_0$ with $|\beta| < 1$, the function f_ν of Lemma 3.8 is $f_\nu = 1$ on $]-\infty, 0[$ and $f_\nu = \frac{1-\beta}{1+\beta}$ on $[0, \infty[$. By Proposition 3.9, $F_\nu(X)$ is then a local martingale, where

$$F_\nu(x) = x\, 1_{\{x<0\}} + \frac{1-\beta}{1+\beta}\, x\, 1_{\{x\geqslant 0\}}.$$

The slopes of this scale function satisfy $(1+\beta)F_\nu'(0+) = (1-\beta)F_\nu'(0-)$; this is the transmission condition that appears in Section 5.2.

In this chapter we follow Lejay [9] and Étoré and Martinez [2]. We compute the joint law of X_t and $\hat{L}_t$, deduce the transition density of X, and describe the partial differential equation it solves.

5.1 The law of skew Brownian motion

5.1.1 Construction by random flipping of excursions

We first describe a construction of skew Brownian motion that shows its relation to Brownian motion [see 13, Chapter XII, Exercise 2.16]. Starting from a reflected Brownian motion, we change the sign of each excursion independently, keeping it positive with probability $\frac{1+\beta}{2}$.

Let B be a Brownian motion started at 0, and let $(Y_n)_{n\geqslant 0}$ be a sequence of independent random variables, independent of B, with

$$\mathbb{P}(Y_n = 1) = \frac{1+\beta}{2}, \qquad \mathbb{P}(Y_n = -1) = \frac{1-\beta}{2}.$$

The set $\{t : B_t \neq 0\}$ is a countable union of disjoint open intervals, the excursion intervals of B. They can be enumerated measurably as $(I_n)_{n\geqslant 0}$: for instance, for each k and j only finitely many of them are contained in $[0, k]$ and have length at least 2^{-j}, and these can be listed by their left end-points. Define

$$X_t = Y_n\,|B_t| \quad \text{if } t \in I_n, \qquad X_t = 0 \quad \text{if } B_t = 0.$$

Thus $|X| = |B|$, and each excursion of X is an excursion of $|B|$ given a random sign. The process X is a skew Brownian motion of parameter β started at 0: there is a Brownian motion W such that X satisfies (5.1) with $x = 0$ [4, 9].

The symmetric local time of X is a functional of $|X|$, since $\hat{L}_t = \lim_{\varepsilon\to 0} \frac{1}{2\varepsilon}\int_0^t 1_{\{|X_s|<\varepsilon\}}\,ds$ by the occupation formula. As $|X| = |B|$, we get $\hat{L}_t = L_t^0(B)$.

Remark 5.2. Lejay [9] describes extensions of this "random flipping of excursions" method, which also give a decomposition of the Itô measure of the excursions of X.

5.1.2 Joint law of skew Brownian motion and its local time

We use the notation

$$p_t(z) = \frac{1}{\sqrt{2\pi t}}\exp\Big(-\frac{z^2}{2t}\Big), \qquad q(a,t) = \frac{a}{\sqrt{2\pi t^3}}\exp\Big(-\frac{a^2}{2t}\Big) = \frac{a}{t}\,p_t(a),$$

for $t > 0$, $z \in \mathbb{R}$ and $a > 0$. Recall that $q(a,\cdot)$ is the density of the hitting time T_a of a by a Brownian motion started at 0. Since T_{a+b} is the sum of T_a and of an independent copy of T_b,

$$\int_0^t q(a,s)\,q(b,t-s)\,ds = q(a+b,t), \qquad a,b>0. \tag{5.2}$$

Lemma 5.3. *Let X be a skew Brownian motion started at x, and W' a Brownian motion started at $|x|$. Then the processes $\big(|X_t|, \hat{L}_t\big)_{t\geqslant 0}$ and $\big(|W'_t|, L_t^0(W')\big)_{t\geqslant 0}$ have the same law.*

Proof. Apply the symmetric Itô–Tanaka formula (Theorem 1.23) to $f(z) = |z|$, whose symmetric derivative is sgn with $\operatorname{sgn}(0) = 0$ and with $f'' = 2\delta_0$:

$$|X_t| = |x| + \int_0^t \operatorname{sgn}(X_s)\, dX_s + \hat{L}_t = |x| + \tilde{W}_t + \hat{L}_t, \qquad \tilde{W}_t = \int_0^t \operatorname{sgn}(X_s)\, dW_s.$$

The term $\beta \int_0^t \operatorname{sgn}(X_s)\, d\hat{L}_s$ vanishes because $d\hat{L}$ is carried by $\{s : X_s = 0\}$. Since $\{s : X_s = 0\}$ has Lebesgue measure 0 by the occupation formula, $\langle \tilde{W} \rangle_t = t$, and $\tilde{W}$ is a Brownian motion by Lévy's characterisation. By Skorokhod's lemma (Lemma 2.3), applied to $y = |x| + \tilde{W}$,

$$\hat{L}_t = \sup_{s \leqslant t} \bigl(-|x| - \tilde{W}_s \bigr) \vee 0, \qquad |X_t| = |x| + \tilde{W}_t + \hat{L}_t.$$

The same computation for W' (for which $\hat{L}^0(W') = L^0(W')$) gives $|W'| = |x| + \tilde{W}' + L^0(W')$ with a Brownian motion $\tilde{W}'$, so both pairs are the same functional of a Brownian motion. □

Lemma 5.4. *Let W be a Brownian motion started at 0. For $t > 0$,*

$$\mathbb{P}_0\bigl[|W_t| \in dy\,;\, L_t^0(W) \in dl \bigr] = 2\, q(y + l, t)\, dy\, dl, \qquad y > 0,\ l > 0.$$

Proof. Let $M_t = \sup_{s \leqslant t} W_s$. By Lévy's theorem, which follows from Skorokhod's lemma as in the proof of Lemma 5.3, $(|W_t|, L_t^0(W))$ has the law of $(M_t - W_t, M_t)$. By the reflection principle [6, Section 2.8], (M_t, W_t) has density $\frac{2(2m-w)}{\sqrt{2\pi t^3}} \exp\bigl(-\frac{(2m-w)^2}{2t} \bigr)$ on $\{m > 0,\ w < m\}$. The change of variables $l = m$, $y = m - w$ has Jacobian 1 and gives the result. □

The next lemma is intuitive, but the presence of the local time makes it hard to prove directly from (5.1). The construction by random flipping makes it easy.

Lemma 5.5. *For $t > 0$ and Borel sets $A \subset]0, \infty[$ and $C \subset [0, \infty[$,*

$$\mathbb{P}_0\bigl[X_t \in A\,;\, \hat{L}_t \in C \bigr] = \frac{1+\beta}{2}\, \mathbb{P}_0\bigl[|X_t| \in A\,;\, \hat{L}_t \in C \bigr],$$
$$\mathbb{P}_0\bigl[X_t \in -A\,;\, \hat{L}_t \in C \bigr] = \frac{1-\beta}{2}\, \mathbb{P}_0\bigl[|X_t| \in A\,;\, \hat{L}_t \in C \bigr].$$

Proof. By uniqueness in law, and since $\hat{L}$ is a functional of the path of X, we may take X to be the process constructed above. Since $\mathbb{P}(B_t = 0) = 0$, almost surely t belongs to some excursion interval I_N, where the index N is $\sigma(B)$-measurable. On $\{|B_t| \in A\}$, we have $X_t \in A$ if and only if $Y_N = 1$. Using $|X| = |B|$, $\hat{L}_t = L_t^0(B)$, and the independence of (Y_n) and B,

$$\begin{aligned}
\mathbb{P}\bigl[X_t \in A\,;\, \hat{L}_t \in C \bigr] &= \mathbb{E}\Bigl[1_{\{|B_t| \in A,\ L_t^0(B) \in C\}}\, \mathbb{P}\bigl(Y_N = 1 \,\big|\, \sigma(B) \bigr) \Bigr] \\
&= \mathbb{E}\Bigl[1_{\{|B_t| \in A,\ L_t^0(B) \in C\}} \sum_n 1_{\{N = n\}}\, \mathbb{P}(Y_n = 1) \Bigr] \\
&= \frac{1+\beta}{2}\, \mathbb{P}\bigl[|X_t| \in A\,;\, \hat{L}_t \in C \bigr].
\end{aligned}$$

The second identity is proved in the same way, with $Y_N = -1$. □

From these lemmas we can compute the joint law of $(X_t, \hat{L}_t)$ [2].

Proposition 5.6. *Let $x \geqslant 0$ and $t > 0$. Under $\mathbb{P}_x$, the law of $(X_t, \hat{L}_t)$ is*

$$\begin{aligned}\mathbb{P}_x\Big[X_t \in dy\,;\, \hat{L}_t \in dl\Big] &= 1_{\{y>0,\, l>0\}}\,(1+\beta)\,q(x+y+l,t)\,dy\,dl \\ &\quad + 1_{\{y<0,\, l>0\}}\,(1-\beta)\,q(x-y+l,t)\,dy\,dl \\ &\quad + 1_{\{y>0\}}\Big(p_t(y-x)-p_t(y+x)\Big)\,dy\,\delta_0(dl).\end{aligned}$$

For $x < 0$, the law is obtained by applying this formula to $-X$, which is a skew Brownian motion of parameter $-\beta$ started at $-x$, with the same symmetric local time at 0.

Proof. Step 1: $x = 0$. By Lemmas 5.5, 5.3 and 5.4, for $u > 0$, $y > 0$ and $l > 0$,

$$\mathbb{P}_0\Big[X_u \in \pm dy\,;\, \hat{L}_u \in dl\Big] = \frac{1 \pm \beta}{2}\,\mathbb{P}_0\Big[|W_u| \in dy\,;\, L_u^0(W) \in dl\Big] = (1 \pm \beta)\,q(y+l,u)\,dy\,dl.$$

Moreover $\hat{L}_u > 0$ for every $u > 0$, $\mathbb{P}_0$-almost surely, since $L_u^0(W)$ has the law of $M_u > 0$. This is the formula for $x = 0$.

Step 2: $x > 0$, $l > 0$. Let $T_0 = \inf\{s : X_s = 0\}$. On $[0, T_0]$, $X = x + W$ and $\hat{L} = 0$, so the law of T_0 has density $q(x, \cdot)$. By the strong Markov property at T_0, and since $X_{T_0} = 0$ and $\hat{L}_{T_0+u} = \hat{L}_u^0(X_{T_0+\cdot})$, Step 1 and (5.2) give, for $y > 0$,

$$\begin{aligned}\mathbb{P}_x\Big[X_t \in dy\,;\, \hat{L}_t \in dl\Big] &= \int_0^t q(x,s)\,\mathbb{P}_0\Big[X_{t-s} \in dy\,;\, \hat{L}_{t-s} \in dl\Big]\,ds \\ &= (1+\beta)\int_0^t q(x,s)\,q(y+l,t-s)\,ds\,dy\,dl = (1+\beta)\,q(x+y+l,t)\,dy\,dl,\end{aligned}$$

and in the same way, for $y < 0$, we get $(1-\beta)\,q(x+|y|+l,t)\,dy\,dl$.

Step 3: $x > 0$, $l = 0$. By Step 1, $\hat{L}_{T_0+u} > 0$ for all $u > 0$, so $\{\hat{L}_t = 0\} = \{t \leqslant T_0\}$, and $\mathbb{P}_x(T_0 = t) = 0$. On $\{t < T_0\}$, $X_t = x + W_t > 0$. By the reflection principle,

$$\mathbb{P}_x\Big[X_t \in dy\,;\, \hat{L}_t = 0\Big] = \mathbb{P}_x\Big[x + W_t \in dy\,;\, t < T_0\Big] = 1_{\{y>0\}}\Big(p_t(y-x) - p_t(y+x)\Big)\,dy. \qquad \square$$

Integrating in l, using $\int_0^\infty q(a+l,t)\,dl = p_t(a)$, we obtain the transition density of X.

Proposition 5.7. *For $t > 0$, $x \in \mathbb{R}$ and $y \neq 0$,*

$$\mathbb{P}_x[X_t \in dy] = p_t^\beta(x,y)\,dy, \qquad p_t^\beta(x,y) = p_t(y-x) + \beta\,\mathrm{sgn}(y)\,p_t(|x|+|y|).$$

Proof. Let $x \geqslant 0$. For $y > 0$, Proposition 5.6 gives $(1+\beta)p_t(x+y) + p_t(y-x) - p_t(y+x) = p_t(y-x) + \beta p_t(x+y)$. For $y < 0$, it gives $(1-\beta)p_t(x+|y|)$, and $x + |y| = |y-x|$, so this is $p_t(y-x) - \beta p_t(|x|+|y|)$. For $x < 0$, apply the result to $-X$, which replaces (x, y, β) by $(-x, -y, -\beta)$ and leaves the formula unchanged. $\square$

For $\beta = 0$ this is the Brownian transition density. Since $\int \mathrm{sgn}(y)\,p_t(|x|+|y|)\,dy = 0$, $p_t^\beta(x,\cdot)$ has total mass 1. Taking $x = 0$ gives $\mathbb{P}_0(X_t > 0) = \frac{1+\beta}{2}$, as the construction by random flipping suggests.

5.2 The partial differential equation solved by skew Brownian motion

The drift $\beta\hat{L}^0(X)$ in (5.1) is singular, so the generator of X is not a classical differential operator. Formally it is

$$\mathcal{A} = \frac{1}{2}\frac{d^2}{dx^2} + \beta\,\delta_0\,\frac{d}{dx}.$$

Portenko [12] identified processes of this type through such generalised operators, and developed their links with the partial differential equation that the transition semigroup solves. This link gives several methods to compute the semigroup of skew Brownian motion. As in Chapter 4, the probabilistic solution can be checked directly.

Proposition 5.8. *Let f be bounded and continuous, and let $u(t,x) = \mathbb{E}_x f(X_t)$. Then*

(i) $u \in C^{1,2}\big(]0,\infty[\times(\mathbb{R}\setminus\{0\})\big)$ *and* $\dfrac{\partial u}{\partial t} = \dfrac{1}{2}\dfrac{\partial^2 u}{\partial x^2}$ *on* $]0,\infty[\times(\mathbb{R}\setminus\{0\})$;

(ii) u *is continuous on* $[0,\infty[\times\mathbb{R}$, *and* $u(0,x) = f(x)$;

(iii) for $t>0$, $\frac{\partial u}{\partial x}(t,\cdot)$ *has limits at* 0^+ *and* 0^-, *and*

$$(1+\beta)\frac{\partial u}{\partial x}(t,0^+) = (1-\beta)\frac{\partial u}{\partial x}(t,0^-).$$

Proof. By Proposition 5.7,

$$u(t,x) = \int p_t(y-x)\,f(y)\,dy + \beta\int \operatorname{sgn}(y)\,p_t(|x|+|y|)\,f(y)\,dy.$$

On each half-line $x>0$ and $x<0$, both kernels $(t,x)\mapsto p_t(y-x)$ and $(t,x)\mapsto p_t(|x|+|y|) = p_t(\pm x+|y|)$ are solutions of the heat equation. Differentiation under the integral sign is justified exactly as in Theorem 4.5, which gives (i). The same domination shows that the x-derivatives extend continuously to $x=0^{\pm}$. Using $p_t'(z) = -\frac{z}{t}p_t(z)$, we get

$$\frac{\partial u}{\partial x}(t,0^{\pm}) = \int \frac{y}{t}\,p_t(y)\,f(y)\,dy \mp \beta\int \operatorname{sgn}(y)\,\frac{|y|}{t}\,p_t(y)\,f(y)\,dy = (1\mp\beta)\,a_t,$$

where $a_t = \int \frac{y}{t}\,p_t(y)\,f(y)\,dy$. Hence $(1+\beta)\partial_x u(t,0^+) = (1+\beta)(1-\beta)a_t = (1-\beta)\partial_x u(t,0^-)$, which is (iii).

For (ii), continuity on $]0,\infty[\times\mathbb{R}$ is clear from the formula. As $t\to 0$ and $x\to x_0$, the first term tends to $f(x_0)$ by Theorem 4.4. The second term is bounded by $\|f\|_\infty\int p_t(|x|+|y|)\,dy$, which tends to 0 if $x_0\neq 0$. If $x_0=0$, it equals $\beta\int_0^\infty p_t(|x|+z)\big(f(z)-f(-z)\big)\,dz$, which tends to 0 because $f(z)-f(-z)\to 0$ as $z\to 0$, while the measures $p_t(|x|+z)\,dz$ on $]0,\infty[$ have mass at most $\frac{1}{2}$ and concentrate near $z=0$. □

Thus u solves the heat equation on each side of 0, is continuous across 0, and satisfies the *transmission condition*

$$(1+\beta)\frac{\partial u}{\partial x}(t,0^+) = (1-\beta)\frac{\partial u}{\partial x}(t,0^-).$$

This condition at 0 involves the skewness parameter, and it makes the equation harder to solve than the heat equation. We now explain where it comes from, and then give an equivalent formulation in divergence form.

5.2.1 The transmission condition

Let $\mathcal{D}$ be the set of continuous functions f that are C^2 on $\mathbb{R}\setminus\{0\}$, with f' and f'' bounded and having limits at 0^+ and 0^-. Such an f is a difference of two convex functions, and in the sense of distributions

$$f''(dz) = f''(z)\,dz + \Big(f'(0^+) - f'(0^-)\Big)\,\delta_0(dz),$$

while its symmetric derivative at 0 is $\frac{1}{2}\Big(f'(0^+)+f'(0^-)\Big)$. By the symmetric Itô–Tanaka formula (Theorem 1.23),

$$f(X_t) - f(x) = \int_0^t f'_{\mathrm{s}}(X_s)\,dW_s + \beta\int_0^t f'_{\mathrm{s}}(X_s)\,d\hat{L}_s + \frac{1}{2}\int_{\mathbb{R}} \hat{L}_t^a\, f''(da).$$

Since $d\hat{L}$ is carried by $\{X=0\}$, the second term is $\frac{\beta}{2}\Big(f'(0^+)+f'(0^-)\Big)\hat{L}_t$. By the occupation formula, the last term is $\frac{1}{2}\int_0^t f''(X_s)\,ds + \frac{1}{2}\Big(f'(0^+)-f'(0^-)\Big)\hat{L}_t$. Hence

$$f(X_t) - f(x) - \frac{1}{2}\int_0^t f''(X_s)\,ds = \int_0^t f'_{\mathrm{s}}(X_s)\,dW_s + \frac{1}{2}\Big((1+\beta)f'(0^+) - (1-\beta)f'(0^-)\Big)\hat{L}_t. \quad (5.3)$$

The stochastic integral is a martingale, because f' is bounded. The last term is a continuous process of bounded variation, and $\hat{L}$ is not identically 0 since X reaches 0. Hence the left-hand side of (5.3) is a martingale if and only if

$$(1+\beta)f'(0^+) = (1-\beta)f'(0^-).$$

In this sense the generator of X acts as $f \mapsto \frac{1}{2}f''$ on the functions of $\mathcal{D}$ that satisfy the transmission condition; see Lejay [9] for a precise description of its domain. For $f = F_\nu$, the scale function of Remark 5.1, $f''=0$ on $\mathbb{R}\setminus\{0\}$, and (5.3) reduces to the martingale property of $F_\nu(X)$.

5.2.2 Divergence form

Assume $|\beta|<1$, and let

$$k(x) = \frac{1-\beta}{2}\,1_{\{x<0\}} + \frac{1+\beta}{2}\,1_{\{x>0\}}, \qquad L = \frac{1}{2k(x)}\frac{d}{dx}\Big(k(x)\frac{d}{dx}\Big),$$

with domain $\mathcal{D}(L) = \{\psi \in H^1(\mathbb{R}) : k\psi' \in H^1(\mathbb{R})\}$. A function $\psi \in \mathcal{D}(L)$ is continuous, and so is $k\psi'$; at 0 this is exactly $(1+\beta)\psi'(0^+) = (1-\beta)\psi'(0^-)$. Away from 0, $L\psi = \frac{1}{2}\psi''$. So L is the generator of skew Brownian motion, written in divergence form [9].

The operator L is self-adjoint and nonpositive on $L^2(\mathbb{R}, k\,dx)$: for $\psi, \varphi \in \mathcal{D}(L)$,

$$\int L\psi\,\varphi\,k\,dx = -\frac{1}{2}\int k\,\psi'\varphi'\,dx.$$

It is associated with the Dirichlet form $\mathcal{E}(\psi,\varphi) = \frac{1}{2}\int k\,\psi'\varphi'\,dx$ on $H^1(\mathbb{R})$. Since k is bounded above and below by positive constants, $L^2(\mathbb{R}, k\,dx)$ and $L^2(\mathbb{R}, dx)$ coincide as sets and have equivalent norms, but L is symmetric only for the weighted scalar product. The transmission problem can then be replaced by its weak formulation: for $\phi \in L^2(\mathbb{R}, k\,dx)$, find

$$u \in C\Big(\mathbb{R}_+; L^2(\mathbb{R}, k\,dx)\Big) \cap L^2\Big(\mathbb{R}_+; H^1(\mathbb{R})\Big), \qquad \frac{\partial u}{\partial t} = Lu, \qquad u(0,\cdot) = \phi.$$

The solution is $u(t,\cdot) = e^{tL}\phi$. An advantage of the divergence form is that the Markov property of the associated process follows directly from the theory of Dirichlet forms.

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